Pricing the Risk of Decision Using Variational Inequalities: An American Type Derivatives Example

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"Just as King Midas learned that gold's true measure
lies not in its lustre but in its utility,
so must we discern that risk's proper valuation
resides not in computation alone,
but in the wisdom to know
what we cannot measure"

Aristotle (Adapted)

Abstract

This thesis investigates a mathematical and computational framework for constrained decision-making based on variational inequalities (VIs). VIs provide a unifying formulation for a broad class of constrained optimization and equilibrium problems, including settings involving strategic interactions and coupled constraints. Extensions of the framework can also accommodate dynamic systems and, under appropriate assumptions, problems involving nonsmooth or discontinuous payoffs. Potential application domains include mathematical finance, robotics, multi-agent systems, and operational decision-making, including business strategy and negotiation.

The study begins by examining the theoretical foundations of elliptic and parabolic variational inequalities in functional spaces, with particular emphasis on existence, uniqueness, and regularity results under suitable assumptions. These foundations are then applied to a benchmark problem in financial mathematics: the pricing of American-style put options. This classical problem can be formulated as a parabolic variational inequality, equivalent under appropriate conditions to a free-boundary formulation of the associated pricing problem. It provides a setting in which to investigate established numerical methods, including penalization and projection techniques, alongside machine-learning-based approaches.

A dedicated chapter introduces Physics-Informed Neural Networks (PINNs) as a mesh-free approach to approximating solutions of variational inequalities. Particular attention is given to the numerical implementation of the methods and to the treatment of the inequality and complementarity conditions. The implementation, numerical experiments, and results are documented in the accompanying code repository (see the following note).

Overall, this thesis combines the mathematical analysis of variational inequalities with their numerical application in financial mathematics. It also provides a basis for future research into the use of VIs and PINNs for more complex, high-dimensional decision-making problems involving uncertainty, dynamic interactions, and constraints.

Note on this edition

This document presents the theoretical part of the thesis, covering the theory of variational inequalities (Chapters 1–2), the formulation of the American put option as a parabolic variational inequality (Chapter 3), the numerical methods considered (Chapter 4), and Physics-Informed Neural Networks (Chapter 5).

The computational implementation, numerical results, and their analysis and discussion are provided in the accompanying GitHub repository:

`https://github.com/Dimi-44/BSc-Dissertation`

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Chapter 1

Foundations of Variational Inequalities

1.1 Variational Inequalities in Functional Spaces

Introduction

Variational inequalities (VIs) arise in a wide range of applied and theoretical contexts. In their most general form, the problem consists of finding an element $x^* \in C$, where C is a subset of some function space, such that:

$$\langle F(x^*), x - x^* \rangle \geq 0 \quad \text{for all } x \in C, \tag{1.1}$$

where F is an operator mapping from C into an appropriate dual space.

Initially, one may consider $C \subset \mathbb{R}^d$, leading to a finite-dimensional, computationally tractable setting. However, many problems in areas such as optimal control, partial differential equations (PDEs), and financial mathematics involve infinite-dimensional spaces. This necessitates a broader functional analytic framework that extends beyond the confines of Euclidean spaces.

Generalization to Functional Spaces

Hilbert Spaces A *Hilbert space* H is a complete inner product space, equipped with:

- An inner product $\langle \cdot, \cdot \rangle : H \times H \to \mathbb{R}$,
- An induced norm $\|x\| = \sqrt{\langle x, x \rangle}$,
- Completeness, meaning every Cauchy sequence in H converges to a point in H.

The inner product generalizes the dot product, allowing us to define notions such as angles and lengths in abstract spaces. Completeness ensures the space is

robust enough for analytical techniques, which is crucial for establishing existence and uniqueness results in variational inequality theory. Moreover, Hilbert spaces support important tools such as orthogonal projections and duality mappings, which play a key role in the analysis of VIs.

L^p Spaces To broaden the functional setting, we consider L^p spaces. For a measure space (Ω, μ), the space $L^p(\Omega)$ consists of measurable functions $f : \Omega \to \mathbb{R}$ such that:

$$\|f\|_{L^p} = \left(\int_\Omega |f|^p \, d\mu \right)^{1/p} < \infty, \quad 1 \leq p < \infty.$$

- When $p = 2$, $L^2(\Omega)$ is a Hilbert space with inner product:

$$\langle f, g \rangle = \int_\Omega fg \, d\mu.$$

- For $p \neq 2$, $L^p(\Omega)$ is a Banach space, a complete normed vector space, but it lacks an inner product structure.

The case $p = 2$ is of particular importance because of the Hilbert space structure, making it especially suited for VI problems involving square-integrable functions.

Sobolev Spaces To address problems involving derivatives, particularly those arising from PDEs, we require a space that accounts for generalized differentiability. *Sobolev spaces* $W^{k,p}(\Omega)$ extend L^p spaces by incorporating weak derivatives:

$$W^{k,p}(\Omega) = \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \text{ for } |\alpha| \leq k\}.$$

The weak derivative $D^\alpha u$ is defined through integration by parts:

$$\int_\Omega u D^\alpha \phi \, dx = (-1)^{|\alpha|} \int_\Omega (D^\alpha u) \phi \, dx,$$

for all test functions $\phi \in C_c^\infty(\Omega)$.

A particularly useful special case is when $p = 2$, in which $W^{k,2}(\Omega)$ becomes a Hilbert space denoted $H^k(\Omega)$. For instance, the inner product in $H^1(\Omega)$ is given by:

$$\langle u, v \rangle_{H^1} = \int_\Omega (uv + \nabla u \cdot \nabla v) \, dx.$$

Sobolev spaces are central to modern VI theory, especially in PDE settings, as they allow for solutions that may not be classically differentiable.

Key Tools and Concepts

Projections in Hilbert Spaces A vital concept in VI analysis is the projection onto a convex set. Given a closed convex subset $C \subset H$, the projection $P_C(x)$ of a point $x \in H$ onto C is the unique point in C such that:

$$\langle x - P_C(x), y - P_C(x) \rangle \leq 0 \quad \text{for all } y \in C.$$

This variational characterization of projections enables reformulations of VIs as fixed-point problems.

Duality and the Riesz-Fréchet Theorem In Hilbert spaces, the dual space H^* consists of all continuous linear functionals on H. The Riesz-Fréchet representation theorem asserts that H^* is isometrically isomorphic to H: for each $\varphi \in H^*$, there exists a unique $y \in H$ such that:

$$\varphi(x) = \langle y, x \rangle \quad \text{for all } x \in H.$$

This identification simplifies the treatment of operators in VIs by allowing us to view them as elements within the same space, although in general this need not be the case.

Weak Formulations In Sobolev spaces, we often deal with weak formulations of variational inequalities. Given a convex subset $K \subset H^1(\Omega)$, a weak VI takes the form: find $u \in K$ such that

$$\int_\Omega A(u)(v - u)\, dx \geq 0 \quad \text{for all } v \in K,$$

where A may involve weak derivatives, such as in the operator $A(u) = -\Delta u + f$. This weak formulation is essential for accommodating non-smooth solutions while preserving the necessary mathematical structure.

1.1.1 Existence and Solvability of Variational Inequalities

Having established the generalized domain for variational inequalities (VIs) in Hilbert and Sobolev spaces, we now investigate a central question:

Does a solution x^* to a VI always exist? Under what conditions is it unique?

This leads us through the geometry of constraint sets, properties of nonlinear operators, and culminates in a powerful result known as the **Lions-Stampacchia Theorem**.

Problem Formulation

We study the variational inequality problem of finding $x^* \in C$ such that:

$$\langle F(x^*), x - x^* \rangle \geq 0 \quad \forall x \in C, \tag{1.2}$$

where:

- $C \subset H$ is a nonempty, convex, closed subset of a Hilbert space H,
- $F : C \to H$ is a (possibly nonlinear) operator.

This inequality models a balance condition rather than an exact match, complicating the question of solvability.

Geometric and Analytical Preliminaries

Structure of the Set C To ensure that a solution is confined within C, we usually assume:

- C is **convex**, i.e. for any $x, y \in C$, the segment between them lies in C,
- C is **closed**, so limit points of sequences in C remain in C,
- In finite dimensions, compactness is ensured if C is also **bounded**.

Regularity of the Operator F Beyond geometry, the operator F must exhibit:

- **Hemicontinuity**:

  $$t \mapsto \langle F(x + t(y - x)), z\rangle \text{ is continuous on } [0, 1].$$

  This ensures directional stability, even when F is not fully continuous.

- **Monotonicity**:

  $$\langle F(y) - F(x), y - x\rangle \geq 0 \quad \forall x, y \in C.$$

  This generalizes non-decreasing behavior and is crucial for existence and uniqueness. Uniqueness requires strict monotonicity. The operator is strictly monotone when

  $$\langle F(y) - F(x), y - x\rangle > 0 \quad \forall x, y \in C,\ x \neq y.$$

- **Coercivity** (needed when C is unbounded): there exists $x_0 \in C$ such that

  $$\frac{\langle F(x) - F(x_0), x - x_0\rangle}{\|x - x_0\|} \to \infty \quad \text{as } \|x\| \to \infty,\ x \in C.$$

  If $0 \in C$, one may take $x_0 = 0$, i.e. $\langle F(x), x\rangle / \|x\| \to \infty$. This replaces compactness and ensures that solutions do not "escape to infinity."

Illustrative Example: Hemicontinuity Without Continuity

Let $H = \mathbb{R}^2$ and define $F(x_1, x_2) = (g(x_1, x_2), 0)$, where

$$g(x_1, x_2) = \begin{cases} \dfrac{x_1 x_2^2}{x_1^2 + x_2^4}, & (x_1, x_2) \neq (0, 0), \\ 0, & (x_1, x_2) = (0, 0). \end{cases}$$

Then:

- F is **hemicontinuous**: g is continuous away from the origin, and on every straight line through the origin it is continuous as well (on the line $x_2 = mx_1$ we have $g = m^2 x_1/(1 + m^4 x_1^2) \to 0$, and $g = 0$ on the line $x_1 = 0$). Hence $t \mapsto \langle F(x + t(y - x)), z\rangle$ is continuous for all x, y, z.
- F is **discontinuous** at the origin: on the parabola $x_1 = x_2^2$ we have $g = 1/2$ for every $x_2 \neq 0$, while $g(0, 0) = 0$.

This shows hemicontinuity is sufficient for many results even when full continuity fails.

Existence in Finite Dimensions

In $\mathbb{R}^n$, the following ensure the existence of solutions to VI problems:

- C is convex, closed, and bounded (i.e., compact),
- F is continuous and monotone.

Example 1.1.1. *Let $C = [0, 1]$, $F(x) = x$. Then the VI becomes:*

$$x^*(x - x^*) \geq 0 \quad \forall x \in [0, 1],$$

which is only satisfied by $x^* = 0$. Since F is strictly monotone, the solution is unique.

Infinite-Dimensional Settings: Challenges and Tools

In infinite-dimensional Hilbert spaces:

- **Compactness** fails: bounded and closed sets are no longer compact.
- **Weak convergence** In order to regain compactness for bounded and closed sets we must change the topology and revert to the weak topology. In this topology the weak convergence:

$$x_n \rightharpoonup x \iff \langle x_n, y\rangle \to \langle x, y\rangle \quad \forall y \in H.$$

To recover existence results, we invoke coercivity of F, ensuring unbounded solutions become "energetically impossible."

The Lions-Stampacchia Theorem

Theorem 1.1.1 (Lions–Stampacchia)**.** *Let H be a real Hilbert space and let $K \subset H$ be a nonempty, closed, convex set. Let $a : H \times H \to \mathbb{R}$ be a continuous bilinear form for which there exist constants $M > 0$ and $\alpha > 0$ satisfying*

$$|a(u,v)| \leq M \|u\|_H \|v\|_H \quad \text{and} \quad a(v,v) \geq \alpha \|v\|_H^2 \qquad \text{for all } u, v \in H.$$

For every $f \in H^*$ there exists a unique $u \in K$ such that

$$a(u, v-u) \geq \langle f, v-u \rangle_{H^*,H} \qquad \text{for all } v \in K. \tag{1.3}$$

Proposition 1.1.2 (Existence for monotone operators)**.** *Let H be a real Hilbert space, let $K \subset H$ be nonempty, closed, and convex, and let $F : K \to H^*$ be monotone and hemicontinuous. Suppose that there is an $x_0 \in C$ such that*

$$\frac{\langle F(x), x - x_0 \rangle_{H^*,H}}{\|x - x_0\|_H} \longrightarrow +\infty \quad \text{as } \|x - x_0\|_H \longrightarrow \infty, \quad x \in C.$$

Then there exists $x^* \in C$ such that

$$\langle F(x^*), y - x^* \rangle_{H^*,H} \geq 0 \qquad \text{for all } y \in C.$$

If F is strictly monotone, the solution is unique.

This nonlinear monotone-operator result is distinct from the classical Lions–Stampacchia theorem for coercive bilinear forms. Kinderlehrer and Stampacchia treat the abstract existence theory in *An Introduction to Variational Inequalities and Their Applications*, Chapter III, Section 1, "An Abstract Existence Theorem," beginning on p. 83[3]. The proof in the appendix separates the standard bounded-set existence result cited from that chapter from the coercivity reduction needed when K is unbounded.

1.1.2 Analysis of Smooth Solutions to Variational Inequalities

Introduction

Having established the functional spaces in which variational inequalities reside and the conditions ensuring their solutions exist, the discussion now turns to the nature of these solutions, beginning with an analysis aimed at understanding their properties. Among the spectrum of solution types, smooth solutions are particularly valuable, as they:

- enhance interpretability, allowing classical derivatives to be meaningfully defined,
- and facilitate numerical approximation, especially when higher-order methods are involved.

However, smoothness is not inherent to all solutions of VIs. It hinges on the problem's structure, notably the properties of the operator involved and the regularity of the domain and data. This naturally leads to two foundational tools in the theory:

- **Elliptic operators**, which promote regularity via coercivity and stability,
- and the **penalization method**, which bridges constrained problems to smoother, unconstrained approximations.

Regularity and Smoothness

Let $u \in K \subset H^1(\Omega)$ be a solution to a variational inequality, where K is a closed convex subset defined by a constraint, such as $u \geq \psi$. A key analytical question is: under what conditions does u possess higher regularity—specifically, does $u \in H^2(\Omega)$ or even smoother spaces?

Smoothness implies that:

- u has weak first and second derivatives in $L^2(\Omega)$,
- and thus can be interpreted classically almost everywhere in Ω.

This depends on the properties of the operator A, an operator which will help us to achieve some useful properties. Thus the inequality is:

$$\langle A(u), v - u \rangle \geq 0 \quad \forall v \in K,$$

where $A : H^1(\Omega) \to H^{-1}(\Omega)$. The H^{-1} is the dual space of $H^1(\Omega)$, containing continuous linear functionals on $H^1(\Omega)$. It includes rougher objects than $L^2(\Omega)$, such as distributions involving derivatives. Hence, it reflects the problem's differential structure [1].

Elliptic Operators and Regularity

Consider the elliptic operator defined by:

$$A(u) = -\sum_{i,j=1}^{n} \partial_{x_j}(a_{ij}(x)\partial_{x_i}u) + a_0(x)u,$$

where $\Omega \subset \mathbb{R}^n$ is an open, bounded domain, $a_{ij}(x)$ are measurable coefficients (with $a_{ij} = a_{ji}$ for symmetry), and $a_0(x) \geq 0$. The operator is *uniformly elliptic* if there exists $\alpha > 0$ such that for all $x \in \Omega$ and $\xi \in \mathbb{R}^n$,

$$\sum_{i,j=1}^{n} a_{ij}(x)\xi_i\xi_j \geq \alpha|\xi|^2.$$

The Bilinear Form: Definition and Properties To understand the operator A, we introduce the associated *bilinear form*, which arises naturally in the weak formulation of partial differential equations (PDEs) and variational problems.

Definition 1.1.1 (Bilinear Form)**.** *A bilinear form $a : H^1(\Omega) \times H^1(\Omega) \to \mathbb{R}$ is a function that is linear in each argument. For the operator A, the associated bilinear form is:*

$$a(u, v) = \int_\Omega \left(\sum_{i,j=1}^{n} a_{ij}(x) \partial_{x_i} u \partial_{x_j} v + a_0(x) uv \right) dx,$$

where $u, v \in H^1(\Omega)$, the Sobolev space of functions with square-integrable weak first derivatives.

This form has two critical properties, stemming from the uniform ellipticity and boundedness of the coefficients:

- **Boundedness**: The form is bounded if there exists a constant $M > 0$ such that:

  $$|a(u, v)| \leq M \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)} \quad \forall u, v \in H^1(\Omega).$$

  This follows because $a_{ij}(x)$ and $a_0(x)$ are assumed bounded (e.g., $|a_{ij}(x)| \leq M_1$, $a_0(x) \leq M_2$), and the Cauchy-Schwarz inequality applies:

  $$|a(u, v)| \leq \int_\Omega \left| \sum_{i,j=1}^{n} a_{ij}(x) \partial_{x_i} u \partial_{x_j} v \right| dx + \int_\Omega |a_0(x) uv| \, dx \leq M_1 \|\nabla u\|_{L^2} \|\nabla v\|_{L^2} + M_2 \|u\|_{L^2} \|v\|_{L^2},$$

  where $\|u\|_{H^1(\Omega)} = \left(\|u\|^2_{L^2(\Omega)} + \|\nabla u\|^2_{L^2(\Omega)} \right)^{1/2}$. Thus, M depends on M_1, M_2.

- **Coercivity**: The form is coercive if there exists $\alpha > 0$ such that:

  $$a(u, u) \geq \alpha \|u\|^2_{H^1(\Omega)} \quad \forall u \in H^1(\Omega).$$

  Uniform ellipticity ensures:

  $$a(u, u) = \int_\Omega \left(\sum_{i,j=1}^{n} a_{ij}(x) \partial_{x_i} u \partial_{x_j} u + a_0(x) u^2 \right) dx \geq \int_\Omega \alpha |\nabla u|^2 \, dx + \int_\Omega a_0(x) u^2 \, dx.$$

  Since $a_0(x) \geq 0$, and the ellipticity condition gives $\sum_{i,j=1}^{n} a_{ij}(x) \partial_{x_i} u \partial_{x_j} u \geq \alpha |\nabla u|^2$, we have:

  $$a(u, u) \geq \alpha \|\nabla u\|^2_{L^2(\Omega)}.$$

  If $a_0(x) \geq \beta > 0$, coercivity over the full $H^1(\Omega)$ norm is clear. Otherwise, for $a_0(x) \geq 0$, coercivity may require boundary conditions (e.g., $u = 0$ on $\partial\Omega$) and Poincaré's inequality to control $\|u\|_{L^2}$.

Connection to the Operator A The bilinear form $a(u, v)$ is intimately tied to the operator $A : H^1(\Omega) \to H^{-1}(\Omega)$. Here's how:

- **Weak Formulation**: In PDEs, we often solve $A(u) = f$, where $f \in H^{-1}(\Omega)$, the dual of $H^1(\Omega)$. Multiplying by a test function $v \in H^1(\Omega)$, integrating, and applying integration by parts (assuming $u = 0$ or $\frac{\partial u}{\partial n} = 0$ on $\partial\Omega$) yields the weak form:

$$\int_\Omega A(u)v\,dx = \int_\Omega \left(-\sum_{i,j=1}^{n} \partial_{x_j}(a_{ij}(x)\partial_{x_i}u) + a_0(x)u \right) v\,dx.$$

  Integration by parts on the first term gives:

$$\int_\Omega -\sum_{i,j=1}^{n} \partial_{x_j}(a_{ij}(x)\partial_{x_i}u)v\,dx = \int_\Omega \sum_{i,j=1}^{n} a_{ij}(x)\partial_{x_i}u\partial_{x_j}v\,dx,$$

  assuming boundary terms vanish. Thus:

$$\int_\Omega A(u)v\,dx = \int_\Omega \sum_{i,j=1}^{n} a_{ij}(x)\partial_{x_i}u\partial_{x_j}v\,dx + \int_\Omega a_0(x)uv\,dx.$$

- **Duality Pairing**: Notice the right-hand side is exactly $a(u, v)$. In the weak sense, we define the operator A via the duality pairing:

$$\langle A(u), v\rangle = a(u, v),$$

  where $\langle\cdot,\cdot\rangle$ denotes the pairing between $H^{-1}(\Omega)$ and $H^1(\Omega)$. This means the operator A acts on $u \in H^1(\Omega)$ to produce a functional $A(u) \in H^{-1}(\Omega)$, evaluated by testing against $v \in H^1(\Omega)$.

- **Implications for Variational Problems**: In variational inequalities or PDEs, we seek $u \in H^1(\Omega)$ (or a convex subset K) such that:

$$\langle A(u), v - u\rangle \geq \langle f, v - u\rangle \quad \forall v \in K,$$

  or, in the PDE case, $a(u, v) = \langle f, v\rangle$. The bilinear form $a(u, v)$ encodes the structure of A, and its coercivity and boundedness (from ellipticity) ensure existence and uniqueness of solutions via theorems like Lax-Milgram.

- **Regularity**: Elliptic regularity relies on these properties. If $f \in H^{-1}(\Omega)$, coercivity of $a(u, u)$ ensures a solution $u \in H^1(\Omega)$. If Ω, $a_{ij}(x)$, and f are smoother (e.g., $f \in L^2(\Omega)$, $a_{ij} \in C^1$, Ω has a smooth boundary), then $u \in H^2(\Omega)$, as the elliptic nature of A "lifts" regularity through $a(u, v)$.

The bilinear form $a(u, v)$ is the bridge between the differential operator A and the weak formulation of PDEs or variational inequalities [1]. Its boundedness and coercivity, derived from the uniform ellipticity of A, are key to proving existence, uniqueness, and regularity of solutions, making it a cornerstone of the analysis.

Penalization Approach

When direct analysis is hindered—due to irregular data or nonsmooth constraints—the penalization technique provides an alternative route. Instead of solving the inequality directly, one considers a family of penalized problems. Consider $C = \{u \geq \psi\}$:

$$A(u_\varepsilon) + \frac{1}{\varepsilon}\beta(u_\varepsilon - \psi) = 0 \quad \text{in } \Omega,$$

where:

- $\varepsilon > 0$ is a small parameter,
- $\beta(s) = 0$ if $s \geq 0$, and $\beta(s) < 0$ if $s < 0$, e.g., $\beta(s) = -s^-$.

This reformulates the inequality as an equation, easing analysis. As $\varepsilon \to 0$, the solution u_ε converges (under suitable conditions) to the solution u of the original VI. Crucially, each u_ε solves the PDE, often admitting higher regularity:

- Classical PDE tools can be applied to each u_ε,
- Regularity results for u_ε can be transferred to u in the limit.

We should note that other choices of C, requires different penalty forms.

Regularity Proof for Elliptic Variational Inequalities

We can prove that under suitable conditions, the solution $u \in K \subset H^1(\Omega)$ to an elliptic variational inequality belongs to $H^2(\Omega)$. This enhances interpretability and supports numerical methods, as discussed in Section 1.1.2 under "Elliptic Operators and Regularity" (p. 10) and "Penalization Approach" (p. 13).

Consider the variational inequality: find $u \in K = \{v \in H_0^1(\Omega) \mid v \geq \psi \text{ a.e. in } \Omega\}$, where $\psi \in H^1(\Omega)$ with $\psi \leq 0$ on $\partial\Omega$ (so that K is nonempty), such that

$$\langle A(u), v - u\rangle \geq \langle f, v - u\rangle \quad \forall v \in K, \tag{1.4}$$

where $A : H^1(\Omega) \to H^{-1}(\Omega)$ is a uniformly elliptic operator defined by

$$A(u) = -\sum_{i,j=1}^{n} \partial_{x_j}(a_{ij}(x)\partial_{x_i} u) + a_0(x)u,$$

with $a_{ij} \in C^{0,1}(\overline{\Omega})$ (Lipschitz continuous), $a_0 \in L^\infty(\Omega)$, $a_0 \geq 0$, and uniform ellipticity: there exists $\alpha > 0$ such that

$$\sum_{i,j=1}^{n} a_{ij}(x)\xi_i\xi_j \geq \alpha|\xi|^2 \quad \forall \xi \in \mathbb{R}^n, \text{ a.e. } x \in \Omega.$$

Assume $f \in L^2(\Omega)$, $\Omega \subset \mathbb{R}^n$ has a smooth boundary, and $\psi \in H^2(\Omega)$.

Theorem 1.1.3. *Let $u \in K \subset H^1(\Omega)$ be the solution to* (1.4)*. If A is uniformly elliptic, $f \in L^2(\Omega)$, $\psi \in H^2(\Omega)$, and $\partial\Omega$ is $C^{1,1}$, then $u \in H^2(\Omega)$.*

Proof. We use the penalization method to prove H^2-regularity. For $\varepsilon > 0$, consider the penalized problem:

$$A(u_\varepsilon) + \frac{1}{\varepsilon}\beta(u_\varepsilon - \psi) = f \quad \text{in } \Omega, \tag{1.5}$$

with $u_\varepsilon \in H^1(\Omega)$, Dirichlet boundary condition $u_\varepsilon = 0$ on $\partial\Omega$, and $\beta(s) = -s^- = \min(s, 0)$. The weak form is: find $u_\varepsilon \in H_0^1(\Omega)$ such that

$$\int_\Omega \sum_{i,j=1}^n a_{ij}\partial_{x_i} u_\varepsilon \partial_{x_j} v + a_0 u_\varepsilon v \, dx + \frac{1}{\varepsilon}\int_\Omega \beta(u_\varepsilon - \psi)v \, dx = \int_\Omega fv \, dx \quad \forall v \in H_0^1(\Omega).$$

Step 1: Existence and regularity of u_ε. The operator A is coercive and bounded (Section 1.1.2), and β is monotone and Lipschitz. By standard elliptic PDE theory, (1.5) has a unique solution $u_\varepsilon \in H_0^1(\Omega)$. Since $f \in L^2(\Omega)$, $\psi \in H^2(\Omega)$, and $\beta(u_\varepsilon - \psi) \in L^2(\Omega)$ (as β is bounded by $|u_\varepsilon - \psi|$), elliptic regularity theory implies $u_\varepsilon \in H^2(\Omega)$, provided $\partial\Omega$ is $C^{1,1}$ and the a_{ij} are Lipschitz. The H^2-norm estimate is uniform in ε. Indeed, since $\psi \le 0 = u_\varepsilon$ on $\partial\Omega$, the function $\beta(u_\varepsilon - \psi) \le 0$ belongs to $H_0^1(\Omega)$; testing (1.5) with it and using $a(w, \min(w, 0)) = a(\min(w, 0), \min(w, 0)) \ge 0$ for $w = u_\varepsilon - \psi$ gives

$$\frac{1}{\varepsilon}\|\beta(u_\varepsilon - \psi)\|_{L^2(\Omega)} \le \|(f - A\psi)^-\|_{L^2(\Omega)},$$

where $A\psi \in L^2(\Omega)$ because $\psi \in H^2(\Omega)$ and the a_{ij} are Lipschitz. Elliptic regularity applied to $Au_\varepsilon = f - \frac{1}{\varepsilon}\beta(u_\varepsilon - \psi)$ then yields

$$\|u_\varepsilon\|_{H^2(\Omega)} \le C\left(\|f\|_{L^2(\Omega)} + \|\psi\|_{H^2(\Omega)}\right),$$

where C is independent of ε.

Step 2: Convergence as $\varepsilon \to 0$. As $\varepsilon \to 0$, $u_\varepsilon \to u$ in $H^1(\Omega)$, where u solves (1.4) (Section 1.1.2). Since $\{u_\varepsilon\}$ is bounded in the Hilbert space $H^2(\Omega)$, there exists a subsequence $u_{\varepsilon_k} \to u$ weakly in $H^2(\Omega)$, and, by the compactness of the embedding $H^2(\Omega) \hookrightarrow H^1(\Omega)$, strongly in $H^1(\Omega)$.

Step 3: $u \in H^2(\Omega)$. Weak convergence in $H^2(\Omega)$ implies $u \in H^2(\Omega)$, and the limit satisfies the VI (1.4). Since the limit is unique (by strict monotonicity of A), the entire sequence $u_\varepsilon \to u$ weakly in $H^2(\Omega)$. Thus, $u \in H^2(\Omega)$.

Step 4: Norm estimate. The uniform bound on $\|u_\varepsilon\|_{H^2(\Omega)}$ and weak convergence imply

$$\|u\|_{H^2(\Omega)} \le \liminf_{\varepsilon\to 0} \|u_\varepsilon\|_{H^2(\Omega)} \le C\left(\|f\|_{L^2(\Omega)} + \|\psi\|_{H^2(\Omega)}\right).$$

Hence, $u \in H^2(\Omega)$, completing the proof. □

This result ensures that the solution u has second-order weak derivatives in $L^2(\Omega)$, enabling classical interpretation almost everywhere and supporting numerical approximations, as discussed in Section 1.1.2.

The Free Boundary in Variational Inequalities

In variational inequalities, particularly those with constraints like $u \geq \psi$, a critical feature is the *free boundary*, the interface separating regions where the solution u is constrained (e.g., $u = \psi$) from regions where it is not (e.g., $u > \psi$). This boundary is "free" because its location is not fixed a priori but determined by the solution itself, making it a key object of study in understanding the behavior and regularity of solutions.

Consider the variational inequality (1.4): find $u \in K = \{v \in H_0^1(\Omega) \mid v \geq \psi \text{ a.e. in } \Omega\}$ such that

$$\langle A(u), v - u \rangle \geq \langle f, v - u \rangle \quad \forall v \in K,$$

where A is the uniformly elliptic operator from Section 1.1.2, $f \in L^2(\Omega)$, and $\psi \in H^2(\Omega)$. The domain Ω splits into two key regions:

- **Coincidence set**: $I = \{x \in \Omega \mid u(x) = \psi(x)\}$, where the constraint is active and u is "pinned" to the obstacle ψ.
- **Non-coincidence set**: $\{x \in \Omega \mid u(x) > \psi(x)\}$, where the constraint is inactive, and u typically satisfies the PDE $A(u) = f$.

The *free boundary* is the interface $F = \partial I \cap \Omega$, the boundary of the coincidence set within Ω, separating these regions.

Properties and Challenges The free boundary is central to applications like obstacle problems, financial mathematics (e.g., American options), and physical systems (e.g., elastic contact). Its properties include:

- **Unknown Location**: Unlike fixed boundaries (e.g., $\partial\Omega$), the free boundary's position depends on u, ψ, f, and A, making it a nonlinear, moving feature.
- **Regularity**: The smoothness of F is tied to the regularity of u, ψ, and f. For example, if $\psi \in H^2(\Omega)$, $f \in L^2(\Omega)$, and $\partial\Omega$ is $C^{1,1}$, Theorem 1.1.3 shows $u \in H^2(\Omega)$, but F may only be continuous or Lipschitz, not necessarily smooth.
- **Behavior**: In the non-coincidence set, u often solves $A(u) = f$, inheriting elliptic regularity (e.g., $u \in H^2(\Omega)$). Near the free boundary, however, u transitions to ψ, and derivatives may be discontinuous or singular, complicating analysis.

Determining the free boundary's location and smoothness is challenging, as it requires understanding where the constraint switches from active to inactive. Classical works, such as Kinderlehrer and Stampacchia [3], establish that under smooth data (e.g., $\psi \in C^2$, $f \in C^0$, smooth Ω), the free boundary may be a C^1 or even analytic surface in some cases, but singularities (e.g., corners, cusps) can occur.

Analysis via Penalization The penalization approach (Section 1.1.2) aids in studying the free boundary. Recall the penalized problem:

$$A(u_\varepsilon) + \frac{1}{\varepsilon}\beta(u_\varepsilon - \psi) = f \quad \text{in } \Omega,$$

where $\beta(s) = -s^- = \min(s, 0)$. As $\varepsilon \to 0$, $u_\varepsilon \to u$ in $H^1(\Omega)$. The term $\frac{1}{\varepsilon}\beta(u_\varepsilon - \psi)$ is large and negative where $u_\varepsilon < \psi$, forcing $u_\varepsilon \approx \psi$ in the coincidence set. We can approximate the free boundary by examining:

- **Approximate Coincidence Set**: $I_\varepsilon = \{x \in \Omega \mid u_\varepsilon(x) \leq \psi(x) + \varepsilon^\gamma\}$, for small $\gamma > 0$, where $u_\varepsilon \approx \psi$.
- **Approximate Free Boundary**: The boundary $\partial I_\varepsilon \cap \Omega$ approximates F. As $\varepsilon \to 0$, this converges (in a suitable sense) to the true free boundary.

Regularity of F depends on the data: if $a_{ij} \in C^1(\Omega)$, $f \in C^0(\Omega)$, and $\psi \in C^2(\Omega)$, results from Brezis [5] and Kinderlehrer and Stampacchia [3] suggest F may be a smooth hypersurface in $\mathbb{R}^n$, though singularities arise if the data lacks smoothness or if the constraint is degenerate.

Relevance to Applications In the context of this thesis, the free boundary is critical for American option pricing, where u represents the option value, ψ the payoff, and the free boundary marks the exercise boundary—where it becomes optimal to exercise the option. The penalization method approximates this boundary, and regularity results help numerical methods locate it accurately. Detailed treatments of free boundary problems in VIs are found in Kinderlehrer and Stampacchia [3] and Tsekrekos and Yannacopoulos [1], with applications to finance explored in the latter.

The pursuit of smooth solutions to variational inequalities is driven by both theoretical elegance and practical necessity. Elliptic operators establish the structural groundwork for regularity, while penalization techniques allow us to circumvent direct difficulties by accessing smoother approximations. Together, they form a robust analytical framework for understanding the fine properties of variational inequality solutions, particularly the behavior of the free boundary critical to applications like American option pricing.

1.1.3 Conclusion

The framework established in this section provides a comprehensive foundation for variational inequalities in infinite-dimensional settings. By generalizing from finite-dimensional Euclidean spaces to Hilbert, L^p, and Sobolev spaces, we construct a flexible analytical structure capable of modeling complex systems governed by PDEs, control constraints, and irregular data [5]. The Lions–Stampacchia theorem gives existence and uniqueness for coercive bilinear forms, and Proposition 1 extends existence to monotone, hemicontinuous and coercive operators, with uniqueness under strict monotonicity [2, 3]. The pursuit of smooth solutions, facilitated by elliptic operators and penalization methods, enhances interpretability and numerical tractability, with the free boundary emerging as a critical feature in applications such as American option pricing [3, 1]. This foundation sets the stage for exploring parabolic VIs, viscosity solutions, computational methods, and their application to financial derivatives.

Chapter 2

Parabolic and Non-Smooth Variational Inequalities

This section extends the variational inequality framework to dynamic and non-smooth settings, addressing time-dependent phenomena and solutions lacking classical differentiability. Parabolic variational inequalities incorporate temporal evolution, modeling systems like optimal control and financial option pricing, while viscosity solutions provide a robust approach to handle non-smooth solutions, particularly at constraint boundaries, ensuring theoretical consistency for complex problems.

2.1 Parabolic Variational Inequalities

Introduction

We now extend our analysis by introducing the dimension of time, which fundamentally alters the nature of the problem. This shift leads us to the concept of **parabolic variational inequalities** (PVIs), a class of variational inequalities that govern time-dependent phenomena, capturing the evolution of solutions over a temporal interval. These problems naturally arise in a variety of contexts where decisions or states evolve dynamically, such as:

- Optimal control problems,
- Heat diffusion with state constraints,
- Financial option pricing.

In these settings, the formulation of the problem integrates both spatial and temporal derivatives within a variational framework. The introduction of time demands a reevaluation of the mathematical structures we have previously relied upon—specifically, the function spaces, operators, and analytical techniques.

These must be adapted to reflect the evolving relationship between past, present, and future states of the system.

Formulation of the Problem

A parabolic variational inequality is posed over a time interval $[0, T]$ and a spatial domain Ω, with the goal of finding a function $u : [0, T] \times \Omega \to \mathbb{R}$ that satisfies a time-dependent inequality while remaining within a convex constraint set K.

Formally, we seek a function $u(t, x) \in K$ such that for almost all $t \in [0, T]$ and all $v \in K$:

$$\langle \partial_t u(t) + Au(t) - f(t), v - u(t) \rangle \geq 0, \tag{2.1}$$

subject to the initial condition $u(0, x) = u_0(x)$ in Ω.

Here:

- $\partial_t u = \frac{\partial u}{\partial t}$ denotes the time derivative,
- A is a spatial operator, typically second-order elliptic,
- $f(t)$ is a given source term in the dual space,
- K is a closed convex subset of a Sobolev space, e.g., $K = \{v \in H^1(\Omega) \mid v \geq \psi(t, x)\}$, where ψ is an evolving obstacle function.

A common choice for A is:

$$Au = -\sum_{i,j} \partial_{x_j}(a_{ij} \partial_{x_i} u) + a_0 u,$$

which ensures coercivity and spatial regularization.

This formulation generalizes static variational inequalities by requiring the inequality to hold at each time t, effectively describing a continuous trajectory $u(t)$ through a constrained function space.

Lebesgue-Bochner and Sobolev-Bochner Spaces

To rigorously handle the time-dependent nature of parabolic variational inequalities, we introduce Lebesgue-Bochner and Sobolev-Bochner spaces. These spaces provide the appropriate functional framework for solutions that evolve over time while satisfying spatial constraints, bridging the temporal and spatial components of the problem.

Lebesgue-Bochner Spaces Let X be a Banach space (e.g., $L^2(\Omega)$ or $H^1(\Omega)$) with norm $\| \cdot \|_X$, and let $[0, T]$ be a time interval. The Lebesgue-Bochner space $L^p(0, T; X)$, for $1 \leq p < \infty$, consists of measurable functions $u : [0, T] \to X$ such that:

$$\|u\|_{L^p(0,T;X)} = \left(\int_0^T \|u(t)\|_X^p \, dt \right)^{1/p} < \infty.$$

For $p = \infty$, the space $L^\infty(0, T; X)$ includes essentially bounded functions with:

$$\|u\|_{L^\infty(0,T;X)} = \operatorname*{ess\,sup}_{t\in[0,T]} \|u(t)\|_X < \infty.$$

Key properties include:

- If X is a Banach space, $L^p(0, T; X)$ is a Banach space for $1 \leq p \leq \infty$.
- For $X = H^1(\Omega)$, $L^2(0, T; H^1(\Omega))$ contains functions $u(t, x)$ that are square-integrable in time with $u(t) \in H^1(\Omega)$ for almost all t, crucial for weak formulations of PVIs.

In the context of PVIs, $L^2(0, T; H^1(\Omega))$ is used to ensure that solutions $u(t)$ have sufficient spatial regularity over time, while $L^2(0, T; H^{-1}(\Omega))$ accommodates weak time derivatives.

Sobolev-Bochner Spaces For functions with time derivatives, we define Sobolev-Bochner spaces. The space $W^{1,p}(0, T; X)$ consists of functions $u \in L^p(0, T; X)$ whose weak time derivative $\partial_t u$ exists and belongs to $L^p(0, T; X)$, with norm:

$$\|u\|_{W^{1,p}(0,T;X)} = \left(\|u\|^p_{L^p(0,T;X)} + \|\partial_t u\|^p_{L^p(0,T;X)}\right)^{1/p}.$$

The weak time derivative $\partial_t u$ is defined such that for all test functions $\phi \in C_c^\infty(0, T)$:

$$\int_0^T u(t)\phi'(t)\,dt = -\int_0^T \partial_t u(t)\phi(t)\,dt,$$

where the integrals are understood in the sense of Bochner integrals. For PVIs, a common space is:

$$H^1(0, T; L^2(\Omega)) = W^{1,2}(0, T; L^2(\Omega)),$$

which includes functions u such that $u, \partial_t u \in L^2(0, T; L^2(\Omega))$. This space is a Hilbert space with inner product:

$$\langle u, v\rangle_{H^1(0,T;L^2(\Omega))} = \int_0^T \langle u(t), v(t)\rangle_{L^2(\Omega)} + \langle \partial_t u(t), \partial_t v(t)\rangle_{L^2(\Omega)}\,dt.$$

For PVIs, solutions are often sought in:

$$u \in L^2(0, T; H^1(\Omega)) \cap H^1(0, T; L^2(\Omega)),$$

ensuring spatial regularity ($u(t) \in H^1(\Omega)$) and temporal regularity ($\partial_t u \in L^2(0, T; L^2(\Omega))$).

Relevance to Parabolic VIs Lebesgue-Bochner and Sobolev-Bochner spaces are essential for the weak formulation of PVIs. The space $L^2(0,T;H^1(\Omega))$ ensures that $u(t) \in K \subset H^1(\Omega)$ for almost all t, satisfying spatial constraints (e.g., $u \geq \psi$). The space $H^1(0,T;L^2(\Omega))$ guarantees that the time derivative $\partial_t u$ is well-defined in a weak sense, allowing the VI to capture temporal evolution. These spaces enable the use of compactness results, such as the Aubin-Lions Lemma (Section 2.1), to prove existence and regularity of solutions.

Weak Formulation

Because the solution u may not be classically differentiable in time, we adopt a weak (variational) formulation. We define:

- $L^2(0,T;H^1(\Omega))$: functions square-integrable in time with values in $H^1(\Omega)$,
- $L^2(0,T;H^{-1}(\Omega))$: space for the weak time derivative.

We seek $u \in L^2(0,T;H^1(\Omega))$ such that:

- $u(t) \in K$ for almost every $t \in [0,T]$,
- $\partial_t u \in L^2(0,T;H^{-1}(\Omega))$,
- $u(0) = u_0$.

The weak form of the VI becomes: for all $v \in L^2(0,T;H^1(\Omega))$ with $v(t) \in K$,

$$\int_0^T \langle \partial_t u(t), v(t) - u(t) \rangle + a(u(t), v(t) - u(t))\, dt \geq \int_0^T \langle f(t), v(t) - u(t) \rangle\, dt,$$

where the bilinear form $a(u,v)$ is defined as:

$$a(u,v) = \int_\Omega \sum_{i,j} a_{ij} \partial_{x_i} u\, \partial_{x_j} v + a_0 uv\, dx.$$

This weak formulation allows for discontinuities or irregularities in u while maintaining the variational structure of the problem. Integration by parts in time ensures a consistent definition of the weak derivative.

Existence of Solutions

Existence of solutions to parabolic variational inequalities can be shown using a generalization of methods from the elliptic (static) case, with added tools to handle time dependence.

Key Assumptions

- The operator A is bounded and coercive,
- The set K is closed and convex,
- The source term $f \in L^2(0,T;H^{-1}(\Omega))$,
- The initial data $u_0 \in L^2(\Omega)$.

Compactness and the Aubin-Lions Lemma A crucial ingredient in the existence proof is the **Aubin-Lions Lemma**, which is a key result in functional analysis used to establish compactness of a sequence of functions in certain Sobolev spaces. Before we delve into the lemma itself, we first need to understand the concept of **compactness** in this context.

What is Compactness? In functional analysis, **compactness** refers to the property that a bounded sequence of functions (or operators) has a subsequence that converges in the appropriate topology in a given space. More precisely, a set S in a Banach space is compact if every sequence in S has a subsequence that converges to an element of S. In simpler terms, it means that a set is "tight" and does not allow for the functions or elements to "escape" in an uncontrolled manner. Compactness is critical for proving the existence of solutions to partial differential equations, as it allows us to pass to the limit in a sequence of approximated solutions.

In the context of variational inequalities, we often need to prove that the sequence of approximate solutions (generated by discretization methods or penalization techniques) has a convergent subsequence. This is where the Aubin-Lions Lemma plays a key role.

The Aubin-Lions Lemma provides a criterion for the compactness of a sequence of functions, specifically in the context of time-dependent problems where both spatial and temporal aspects must be considered. The lemma states that:

Theorem 2.1.1 (Aubin-Lions Lemma)**.** *Let X, Y, and Z be Banach spaces, and let $S = (0, T)$ be a bounded time interval. Suppose that the following conditions hold:*

- *$X \hookrightarrow Y \hookrightarrow Z$, where the embedding of X into Y is compact and the embedding of Y into Z is continuous*[1]*,*
- *A sequence $\{u_n\}$ is bounded in $L^p(S; X)$ for some $1 \leq p < \infty$,*
- *The sequence $\{\partial_t u_n\}$ is bounded in $L^q(S; Z)$ for some $1 \leq q \leq \infty$.*

Then, the sequence $\{u_n\}$ is relatively compact in $L^p(S; Y)$ [10].

This result allows us to extract a convergent subsequence from a bounded sequence of functions, which is essential for proving the existence of weak solutions to parabolic variational inequalities. The sequence $\{u_n\}$ represents a sequence of approximate solutions, and the Aubin-Lions Lemma ensures that we can find a convergent subsequence in the appropriate function space. This compactness property is key to showing that the weak limit of the subsequence is indeed a solution to the original variational inequality.

[1] A continuous embedding is a one-to-one mapping that is also continuous; a compact embedding maps bounded sets to relatively compact sets.

Finally Using the compactness result from the Aubin-Lions Lemma, we can conclude that a weak solution to the parabolic variational inequality exists. The compactness ensures that the approximated solutions converge to a valid solution, completing the existence proof.

Penalization Method

To approximate the VI, we use the penalization method, which transforms the inequality into a nonlinear PDE with a penalization term.

Penalized Problem Let $K = \{v \in H^1(\Omega) \mid v \geq \psi(t,x)\}$. For $\varepsilon > 0$, we solve:

$$\partial_t u_\varepsilon + Au_\varepsilon + \frac{1}{\varepsilon}\beta(u_\varepsilon - \psi) = f,$$

with initial condition $u_\varepsilon(0) = u_0$, where $\beta(s) = -s^-$ penalizes violations of the constraint $u \geq \psi$.

Convergence As $\varepsilon \to 0$, the solution u_ε converges (in the weak topology of $L^2(0,T;H^1(\Omega))$) to a solution u of the original VI. The penalization approach not only provides existence by approximation but also facilitates numerical schemes and regularity analysis.

Regularity of Solutions

The interplay between the elliptic operator A and the time derivative ∂_t provides smoothing in both space and time—though the constraint $u \in K$ can limit this regularity near the boundary of K.

Improved Regularity Under the assumptions:

- A is uniformly elliptic,
- $f \in L^2(0,T;L^2(\Omega))$,
- $\psi \in H^1(0,T;H^1(\Omega))$,
- Ω has a smooth boundary,

the solution u belongs to:

$$H^1(0,T;L^2(\Omega)) \cap L^2(0,T;H^2(\Omega)).$$

This means that:

- The time derivative $\partial_t u \in L^2(0,T;L^2(\Omega))$,
- The spatial regularity improves to $u(t) \in H^2(\Omega)$ for almost all t.

2.1.1 Viscosity Solutions

Introduction

After examining the regularity of solutions to variational inequalities (VIs) in both static and time-dependent settings—where smoothness emerges under specific conditions tied to elliptic and parabolic operators—we now confront a critical scenario: the case where solutions fail to exhibit classical smoothness.

- This lack of regularity often arises at boundaries or interfaces defined by the constraint set.
- It challenges the direct application of differential calculus and necessitates an alternative framework.

We turn our attention to **viscosity solutions**, a powerful concept designed to handle non-smooth solutions rigorously. Viscosity solutions extend the reach of VIs by providing a weak notion of solution that applies even when classical derivatives are unavailable, ensuring theoretical consistency across a broader class of problems. They are also an interesting alternative to the concept of weak topology introduced above, in the case where the relevant equations involve a fully nonlinear structure.

Challenges of Non-Smooth Solutions

The discussion of non-smooth solutions begins with the recognition that VIs, whether elliptic or parabolic, frequently produce solutions that are only piecewise smooth or continuous, particularly near the boundary of the constraint set K.

Example: Variational Inequality Formulation For instance, consider the following VI formulations:

- **Elliptic case:**
$$\langle A(u), v - u \rangle \geq 0 \quad \text{for all } v \in K.$$
- **Parabolic case:**
$$\langle \partial_t u + Au - f, v - u \rangle \geq 0.$$

The solution u may coincide with an obstacle ψ in some regions and deviate in others, leading to discontinuities in derivatives at the transition. Classical solutions, requiring u to be differentiable and satisfy the inequality pointwise, become untenable in such cases.

Viscosity Solutions as a Generalized Approach To address this, viscosity solutions offer a generalized approach:

- They redefine what it means for u to solve the VI by relying on test functions rather than the solution's own differentiability.
- This captures the solution's behavior in a weaker, yet mathematically precise, sense.

Formal Definition of Viscosity Solutions

A viscosity solution to a variational inequality is defined in the context of a nonlinear equation derived from the VI, such as:

$$F(t, x, u, \partial_t u, \nabla u, D^2 u) = 0,$$

where F encapsulates the operator and constraint. For a parabolic VI, this is often expressed as:

$$F = \min\{\partial_t u + Au - f, u - \psi\}. \tag{2.2}$$

Subsolution and Supersolution Conditions A function $u : [0, T] \times \Omega \to \mathbb{R}$ is a viscosity solution if it is continuous and satisfies two conditions:

(i) **Viscosity subsolution:** For any smooth test function $\phi \in C^{1,2}([0, T] \times \Omega)$ that touches u from above at a point (t, x) (i.e., $\phi \geq u$ near (t, x) and $\phi(t, x) = u(t, x)$), we require:

$$F(t, x, u(t, x), \partial_t \phi(t, x), \nabla \phi(t, x), D^2 \phi(t, x)) \leq 0,$$

or equivalently:

$$\min\{\partial_t \phi(t, x) + A\phi(t, x) - f(t, x), \phi(t, x) - \psi(t, x)\} \leq 0.$$

(ii) **Viscosity supersolution:** For any smooth test function $\chi \in C^{1,2}([0, T] \times \Omega)$ that touches u from below at (t, x) (i.e., $\chi \leq u$ near (t, x) and $\chi(t, x) = u(t, x)$), we require:

$$F(t, x, u(t, x), \partial_t \chi(t, x), \nabla \chi(t, x), D^2 \chi(t, x)) \geq 0,$$

or:

$$\min\{\partial_t \chi(t, x) + A\chi(t, x) - f(t, x), \chi(t, x) - \psi(t, x)\} \geq 0.$$

A function is a viscosity solution if it is both a sub- and supersolution. This definition sidesteps the need for u to possess derivatives by transferring the differential requirements to the test functions ϕ and χ, making it suitable for nonsmooth solutions [4].

Equivalence to the Variational Inequality Formulation To connect the viscosity solution framework to the variational inequality (1.4) introduced in Chapter 1, consider the weak formulation of the VI: find $u \in K = \{v \in H^1(\Omega) \mid v \geq \psi \text{ a.e. in } \Omega\}$ such that

$$\langle A(u), v - u\rangle \geq \langle f, v - u\rangle \quad \forall v \in K, \tag{2.3}$$

where $A : H^1(\Omega) \to H^{-1}(\Omega)$ is the elliptic operator defined previously, and $f \in H^{-1}(\Omega)$. For parabolic problems, we extend this to include time dependence, seeking $u : [0, T] \times \Omega \to \mathbb{R}$ in a suitable space (e.g., $L^2(0, T; H^1(\Omega))$) satisfying:

$$\int_0^T \langle \partial_t u(t) + Au(t), v(t) - u(t)\rangle \, dt \geq \int_0^T \langle f(t), v(t) - u(t)\rangle \, dt \quad \forall v \in L^2(0, T; K).$$

The viscosity formulation (2.2) is equivalent to (2.3) in the following sense:

- **Weak to Strong Form**: The weak VI (2.3) arises from integrating the strong form of the PDE or inequality against test functions. In regions where $u > \psi$, the constraint is inactive, and (2.3) implies $\partial_t u + Au = f$ (in a weak sense). Where $u = \psi$, the constraint $u \geq \psi$ is active, and the VI enforces $\partial_t u + Au \geq f$. The viscosity formulation (2.2) captures this by combining both cases into a single equation:

$$\min\{\partial_t u + Au - f, u - \psi\} = 0,$$

  meaning either $\partial_t u + Au = f$ (when $u > \psi$) or $u = \psi$ (when the constraint is active).

- **Viscosity Framework**: The viscosity approach handles nonsmooth solutions by using test functions to evaluate F at points of contact. For smooth solutions, the viscosity conditions reduce to the classical PDE or inequality. Specifically, if u is smooth and satisfies (2.2) classically, then:

$$\partial_t u + Au = f \quad \text{where } u > \psi, \quad \text{and} \quad u \geq \psi \text{ everywhere},$$

  with $\partial_t u + Au \geq f$ where $u = \psi$. This is consistent with the weak VI (2.3) when tested against $v \in K$, as shown in [3].

- **Complementarity**: The viscosity formulation can be interpreted as a complementarity problem. Define the contact set $I = \{(t, x) \mid u(t, x) = \psi(t, x)\}$. The VI (2.3) implies:

$$(\partial_t u + Au - f)(u - \psi) = 0,$$

  meaning either $u = \psi$ (in I) or $\partial_t u + Au = f$ (outside I). This is precisely the condition enforced by (2.2), as the min ensures one term is zero at each point [1].

- **Handling Nonsmoothness**: The weak VI (2.3) assumes $u \in H^1(\Omega)$, but solutions may lack higher regularity near the free boundary. The viscosity framework, by relying on test functions, extends the solution concept to continuous but nonsmooth u, aligning with the needs of problems like American option pricing, where the exercise boundary introduces discontinuities in derivatives [4, 1].

Thus, the viscosity formulation (2.2) is a strong-form equivalent of the weak VI (2.3), adapted to handle nonsmooth solutions while preserving the constraint $u \geq \psi$ and the operator's action. This equivalence is detailed in [3] and applied to financial contexts in [1].

Key Properties of Viscosity Solutions

Stability Under Perturbations The strength of viscosity solutions lies in their stability, a property that ensures robustness under perturbations:

- If a sequence of functions u_n converges uniformly to u on $[0,T] \times \Omega$, and each u_n is a viscosity solution to a perturbed VI (e.g., with operators F_n converging to F), then u remains a viscosity solution to the limiting VI under mild continuity assumptions on F.
- This stability is rooted in the pointwise nature of the definition, which relies on local comparisons with smooth test functions rather than global differentiability [4].

Existence via Perron's Method Establishing the existence of viscosity solutions requires a distinct approach, often employing **Perron's method**, which constructs the solution as an envelope of sub- and supersolutions.

For a VI like:

$$\min\{\partial_t u + Au - f, u - \psi\} = 0,$$

we define:

- A **subsolution** w satisfies:

$$\min\{\partial_t w + Aw - f, w - \psi\} \leq 0 \quad \text{(in the viscosity sense)}.$$

- A **supersolution** z satisfies:

$$\min\{\partial_t z + Az - f, z - \psi\} \geq 0 \quad \text{(in the viscosity sense)}.$$

Perron's method defines:

$$u(t,x) = \sup\{w(t,x) \mid w \text{ is a subsolution}, w \leq z \text{ for some supersolution } z\}.$$

Under appropriate conditions (e.g., bounded domain, continuous f, ψ, and suitable boundary conditions), this u is continuous and can be shown to be a viscosity solution [4]. This method is particularly effective for parabolic VIs, as it leverages the comparison principle to ensure uniqueness [3].

Applications: Optimal Stopping Problems

The relevance of viscosity solutions to VIs is particularly pronounced in their connection to **optimal stopping problems**, a subclass of VIs where the solution represents the value of a decision process.

For a parabolic VI of the form:

$$\min\{\partial_t u + Au - f, u - \psi\} = 0,$$

the solution $u(t,x)$ may represent:

- The maximum expected payoff of stopping a process at time t versus continuing.
- ψ as the immediate reward.

- $Au - f$ as the generator of the underlying dynamics.

Viscosity solutions naturally accommodate the **free boundary**—the interface between stopping and continuation regions—even when this boundary is irregular or the solution lacks differentiability.

2.1.2 Conclusion

Parabolic variational inequalities and viscosity solutions together form a comprehensive framework for addressing time-dependent and non-smooth problems in constrained systems. Parabolic VIs extend static VI theory by incorporating temporal evolution, using Lebesgue-Bochner and Sobolev-Bochner spaces to ensure spatial and temporal regularity, with existence secured via the Aubin-Lions Lemma and penalization methods [3]. Their weak formulation and regularity results enable modeling of dynamic processes in optimal control, heat diffusion, and finance [1]. Viscosity solutions address the challenge of non-smooth solutions, particularly at constraint boundaries, by redefining solutions through test functions, offering stability, existence via Perron's method, and applicability to optimal stopping problems like American option pricing [4, 1]. This dual approach ensures robustness across smooth and non-smooth regimes, critical for applications with complex dynamics and irregular boundaries.

Chapter 3

American Options as Variational Inequality Problem

3.1 Pricing American Put Options as a Variational Inequality

The pricing of American put options, which permit early exercise, is a canonical optimal stopping problem that aligns naturally with the variational inequality (VI) framework developed in Chapters 1–2. This section rigorously connects the financial problem to the theoretical foundations of functional spaces, existence and uniqueness, and smooth solutions (Chapter 1), parabolic VIs and viscosity solutions (Chapter 2), and computation (Chapter 4). Each component of the American option VI is anchored to these chapters, ensuring a robust link between theory and application. Additionally, we prove why viscosity solutions are essential for handling non-smoothness at the free boundary, a critical feature of this problem.

3.2 Problem Setup: Optimal Stopping

The valuation of American put options, which permit exercise at any time before expiration at time T, is a classic optimal stopping problem that we formulate as a variational inequality (VI). Let $V(t, Y_t)$ denote the option value at time t with underlying asset price Y_t, and $\psi(Y_t) = \max(K - Y_t, 0)$ the payoff for a strike price K. At expiration, $V(T, Y_T) = \psi(Y_T)$. The holder's decision to exercise for $\psi(Y_t)$ or continue holding the option is modeled as a VI, leveraging the functional spaces and weak formulations introduced in Chapter 1 [1].

The VI formulation arises from the principle of dynamic programming,

specifically Bellman's optimality condition, which ensures the option value is at least the immediate exercise payoff:

$$V(t, Y_t) \geq \psi(Y_t). \tag{3.1}$$

If $V(t, Y_t) < \psi(Y_t)$, exercising would yield a higher payoff, contradicting optimality, since the holder can always exercise. This condition defines the convex set $K = \{v \in H^1(\Omega) \mid v \geq \psi \text{ a.e. in } \Omega\}$, as discussed in Section 1.1.2 (p. 10). The VI requires finding $V \in K$ such that:

$$\langle -\partial_t V + AV - f, v - V \rangle \geq 0 \quad \forall v \in K, \tag{3.2}$$

where A is the Black-Scholes operator (3.11) defined below, an elliptic operator of the type studied in Section 1.1.2 (p. 10), f is a running payoff (source term), with $f = 0$ for the American put, and the Sobolev space $H^1(\Omega)$ accommodates the weak formulation (p. 6). This VI encapsulates the optimal stopping problem: in regions where $V > \psi$, the option is held ($-\partial_t V + AV = f$), and where $V = \psi$, exercising is optimal [3, 1].

The problem's structure yields two regions:

- **Continuation Region**: $\{(t, Y_t) \mid V(t, Y_t) > \psi(Y_t)\}$, where holding the option is optimal, and V satisfies the PDE $-\partial_t V + AV = f$.
- **Exercise Region**: $\{(t, Y_t) \mid V(t, Y_t) = \psi(Y_t)\}$, where exercising yields the payoff $\psi(Y_t)$.

The *free boundary*, located at price Y_t^*, separates these regions and is determined endogenously, making this a free boundary problem (p. 15). The projection operator onto K, defined as $P_K(z) = \arg\min_{y \in K} \|z - y\|^2$ (p. 6), ensures V remains feasible. Since the free boundary may be non-smooth, viscosity solutions, as detailed in Section 3.2.3 and Section 2.1.1 (p. 27), are employed to handle potential irregularities in V [4].

The connection to dynamic programming clarifies the VI's role. Bellman's principle suggests that $V(t, Y_t)$ is the supremum over stopping strategies, balancing immediate exercise ($\psi(Y_t)$) and continuation value. This leads to the dynamic programming equation:

$$V(t, Y_t) = \max\{\psi(Y_t), \mathbb{E}[e^{-r\Delta t} V(t + \Delta t, Y_{t+\Delta t}) \mid Y_t]\},$$

which, in the continuous-time limit, translates to the VI (3.2) for parabolic problems. The VI thus operationalizes dynamic programming by embedding the constraint $V \geq \psi$ and the operator A into a weak formulation, enabling rigorous analysis and computation, as explored in Chapter 4 [1].

3.2.1 Deriving the VI from the Black-Scholes-Merton Framework

Assume Y_t follows a geometric Brownian motion under the risk-neutral measure:

$$dY_t = rY_t dt + \sigma Y_t dW_t, \tag{3.3}$$

where r is the risk-free rate, σ is volatility, and W_t is a Wiener process. In the continuation region ($V > \psi$), the Black-Scholes-Merton (BSM) PDE governs:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 Y_t^2 \frac{\partial^2 V}{\partial Y_t^2} + rY_t \frac{\partial V}{\partial Y_t} - rV = 0. \tag{3.4}$$

This PDE arises from risk-neutral pricing. By Itô's lemma:

$$dV = \left(\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 Y_t^2 \frac{\partial^2 V}{\partial Y_t^2} + rY_t \frac{\partial V}{\partial Y_t} \right) dt + \sigma Y_t \frac{\partial V}{\partial Y_t} dW_t. \tag{3.5}$$

Under risk-neutrality, $\mathbb{E}[dV] = rV\,dt$, and since $\mathbb{E}[dW_t] = 0$, equating drift terms yields the PDE. Define:

$$W = \frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 Y_t^2 \frac{\partial^2 V}{\partial Y_t^2} + rY_t \frac{\partial V}{\partial Y_t} - rV. \tag{3.6}$$

In the continuation region, $W = 0$, $V > \psi$. In the exercise region, $V = \psi$, and typically $W \leq 0$, as the PDE fails for a non-optimal value. The VI is:

$$W \leq 0, \quad V \geq \psi, \tag{3.7}$$

$$\langle W, V - \psi \rangle = 0, \tag{3.8}$$

or:

$$\min\{-W, V - \psi\} = 0. \tag{3.9}$$

The complementarity condition $\langle W, V - \psi \rangle = 0$ ensures exclusivity: $V > \psi$ implies $W = 0$ (continuation), while $W < 0$ implies $V = \psi$ (exercise). This matches the parabolic VI form (2.1) of Chapter 2, where W combines a time derivative and an elliptic operator, solved in $L^2(0,T;H^1(\Omega)) \cap H^1(0,T;L^2(\Omega))$ (p. 19). In terms of the operator A of (3.11) below, $W = \partial_t V - AV$. The operator A is elliptic, of the type studied in Section 1.1.2 (p. 10); its diffusion coefficient $\frac{1}{2}\sigma^2 Y^2$ vanishes at $Y = 0$, so it is degenerate there, and it becomes uniformly elliptic after the change of variable $x = \ln Y$. The classical Lions-Stampacchia theorem gives existence and uniqueness for variational inequalities generated by continuous coercive bilinear forms on a Hilbert space. Its static statement does not, by itself, establish existence for the time-dependent American put variational inequality. The latter requires the appropriate evolution-space assumptions and a parabolic existence result; the present discussion uses the static theorem as background for the elliptic operator and the general variational-inequality framework. The Riesz-Fréchet theorem (p. 6) identifies dual space operators, simplifying $\langle W, V - \psi \rangle$.

3.2.2 Time-to-Expiry Transformation

To align with [1] (Section 1.2.3, p. 21), we use time-to-expiry $q = T - t$, so $V(t, Y_t) = V(q, Y_q)$, and $\frac{\partial V}{\partial t} = -\frac{\partial V}{\partial q}$. The operator becomes:

$$W = -\frac{\partial V}{\partial q} + \frac{1}{2}\sigma^2 Y_q^2 \frac{\partial^2 V}{\partial Y_q^2} + rY_q \frac{\partial V}{\partial Y_q} - rV. \tag{3.10}$$

Define:

$$AV = -\frac{1}{2}\sigma^2 Y_q^2 \frac{\partial^2 V}{\partial Y_q^2} - rY_q \frac{\partial V}{\partial Y_q} + rV, \tag{3.11}$$

so that $W = -\left(\frac{\partial V}{\partial q} + AV\right)$. With this sign, A has the sign convention of Chapters 1–2 (A coercive). We include a running payoff $f(Y_q)$ (typically $f = 0$). The VI is:

$$\frac{\partial V}{\partial q} + AV - f \geq 0, \quad V \geq \psi, \tag{3.12}$$

$$\left\langle \frac{\partial V}{\partial q} + AV - f, V - \psi \right\rangle = 0, \tag{3.13}$$

or:

$$\min\left\{ \frac{\partial V}{\partial q} + AV - f, V - \psi \right\} = 0. \tag{3.14}$$

This form is a parabolic VI of the form (2.1), where $\frac{\partial V}{\partial q} + AV = f$ in the continuation region ($V > \psi$), and $V = \psi$ in the exercise region. The solution space $L^2(0,T;H^1(\Omega)) \cap H^1(0,T;L^2(\Omega))$ (p. 20) handles weak time derivatives, with the Aubin-Lions Lemma (Theorem 2.1.1) ensuring existence via compactness. The penalization method (p. 23) approximates the VI as a PDE, converging to V, and supports regularity analysis (p. 23). If A is uniformly elliptic, $f \in L^2(0,T;L^2(\Omega))$, and $\psi \in H^1(0,T;H^1(\Omega))$, then $V \in H^1(0,T;L^2(\Omega)) \cap L^2(0,T;H^2(\Omega))$ (p. 23). Non-smoothness at the free boundary is addressed by viscosity solutions (p. 27), with numerical solutions enabled by Minty's reformulation (Theorem 4.1.1).

3.2.3 Necessity of Viscosity Solutions

The free boundary in American option pricing often lacks smoothness, necessitating viscosity solutions (Section 2.1.1). Here, we prove why classical or smooth solutions are insufficient, focusing on the parabolic VI $\min\left\{ \frac{\partial V}{\partial q} + AV - f, V - \psi \right\} = 0$.

Consider the VI in the continuation region ($V > \psi$), where:

$$\frac{\partial V}{\partial q} + AV = f, \tag{3.15}$$

and in the exercise region ($V = \psi$), where typically:

$$\frac{\partial V}{\partial q} + AV \geq f. \tag{3.16}$$

At the free boundary $Y_q = Y_q^*$, $V = \psi$, and the solution transitions between regions. For a classical solution, V must be $C^{1,2}$ (continuous first time derivative, second spatial derivatives) across the boundary, satisfying the PDE pointwise.

However, the payoff $\psi(Y_q) = \max(K - Y_q, 0)$ is only piecewise smooth, with a kink at $Y_q = K$. Applying A to $V = \psi$ in the exercise region yields:

$$A\psi = -\frac{1}{2}\sigma^2 Y_q^2 \frac{\partial^2 \psi}{\partial Y_q^2} - rY_q \frac{\partial \psi}{\partial Y_q} + r\psi. \tag{3.17}$$

Since $\psi(Y_q) = K - Y_q$ for $Y_q < K$, we compute:

$$\frac{\partial \psi}{\partial Y_q} = -1, \quad \frac{\partial^2 \psi}{\partial Y_q^2} = 0 \quad (Y_q \neq K), \tag{3.18}$$

so:

$$A\psi = -\frac{1}{2}\sigma^2 Y_q^2 \cdot 0 - rY_q \cdot (-1) + r(K - Y_q) = rY_q + rK - rY_q = rK. \tag{3.19}$$

The time derivative $\frac{\partial \psi}{\partial q} = 0$, so:

$$\frac{\partial \psi}{\partial q} + A\psi = 0 + rK \geq 0 \quad (\text{if } f = 0), \tag{3.20}$$

which holds in the exercise region. However, at $Y_q = K$, $\frac{\partial^2 \psi}{\partial Y_q^2}$ is undefined, and V may not be $C^{1,2}$ across the boundary, as the transition to the continuation region (where AV involves non-zero second derivatives) introduces discontinuities in higher derivatives. Theorem 1.1.3 shows that elliptic regularity requires smooth data, but ψ's kink violates this. The regularity discussion of Chapter 2 (p. 23) notes that constraints like $V \geq \psi$ limit regularity near the boundary.

Viscosity solutions (p. 25) resolve this by defining V via test functions, not requiring differentiability. For a test function $\phi \in C^{1,2}$ touching V from above at (q, Y_q), the VI requires:

$$\min\left\{\frac{\partial \phi}{\partial q} + A\phi - f, \phi - \psi\right\} \leq 0. \tag{3.21}$$

This holds even if V lacks derivatives at the free boundary, as shown on p. 27 for optimal stopping. Perron's method (p. 27) ensures existence, and stability (p. 26) supports numerical approximations. Thus, viscosity solutions are necessary to handle the non-smooth free boundary, completing the theoretical framework.

Chapter 4

Computational Methods for Variational Inequalities

This section bridges the theoretical framework of variational inequalities (VIs) to their practical implementation, focusing on computational strategies for solving VIs and their parabolic counterparts. We explore how Minty's reformulation transforms VIs into computationally tractable forms, enabling fixed-point algorithms, and delve into numerical methods tailored for parabolic VIs, particularly for pricing American put options, using penalization, finite differences, and projection techniques.

4.1 From Theory to Computation

The ultimate goal of applying variational inequalities often involves solving them numerically for applications such as:

- Pricing financial instruments
- Simulating physical systems
- Optimizing decisions

4.1.1 The Computation Challenge

The standard formulation of variational inequalities presents inherent computational difficulties. Consider the VI problem:

$$\text{Find } u \in K \text{ such that } \langle A(u), v - u \rangle \geq 0 \quad \forall v \in K \tag{4.1}$$

where:

- K is a closed convex subset of a Hilbert space H
- $A : H \to H^*$ is an operator (often identified with H via duality)

Computational Limitations Numerical algorithms face several challenges with this formulation:

- The inequality involves testing against an infinite set of directions $v - u$
- Direct computation is infeasible in infinite-dimensional spaces like $H^1(\Omega)$ or $L^2(0, T; H^1(\Omega))$
- Constraint sets K (e.g., $K = \{v \in H^1(\Omega) \mid v \geq \psi\}$) introduce additional complexity

4.1.2 Minty's Reformulation

Minty's reformulation provides an equivalent characterization of the variational inequality (VI) problem, offering a powerful tool for both theoretical analysis and computational methods. For a monotone operator $A : H^1(\Omega) \to H^{-1}(\Omega)$ (satisfying $\langle A(u) - A(v), u - v\rangle \geq 0$), as defined in Section 1.1.1 (p. 7), the reformulation connects the VI to a condition that is particularly useful for deriving fixed-point algorithms, as we explore in Section 4.1.2.

Theorem 4.1.1 (Minty's Reformulation)**.** *Let $K \subset H^1(\Omega)$ be a nonempty, closed, convex subset, and A a monotone, hemicontinuous operator. The following are equivalent:*

1. *$u \in K$ solves the VI:*
$$\langle A(u), v - u\rangle \geq 0 \quad \forall v \in K, \tag{4.1}$$
2. *$u \in K$ satisfies:*
$$\langle A(v), v - u\rangle \geq 0 \quad \forall v \in K. \tag{4.2}$$

Proof. The proof relies on the monotonicity and hemicontinuity of A.

(1) $\implies$ (2): Suppose $u \in K$ solves (4.1), i.e., $\langle A(u), v - u\rangle \geq 0$ for all $v \in K$. Since A is monotone, for any $v \in K$,

$$\langle A(v) - A(u), v - u\rangle \geq 0.$$

Rearranging, we get:

$$\langle A(v), v - u\rangle \geq \langle A(u), v - u\rangle \geq 0,$$

where the second inequality follows from (4.1). Thus, $\langle A(v), v - u\rangle \geq 0$ for all $v \in K$.

(2) $\implies$ (1): Suppose $\langle A(v), v - u\rangle \geq 0$ for all $v \in K$. Consider $v_t = u + t(w - u)$, where $w \in K$, $t \in (0, 1]$, so $v_t \in K$ by convexity of K. Then:

$$\langle A(v_t), v_t - u\rangle \geq 0 \implies \langle A(u + t(w - u)), t(w - u)\rangle \geq 0 \implies \langle A(u + t(w - u)), w - u\rangle \geq 0.$$

Since A is hemicontinuous, as $t \to 0^+$, $A(u + t(w - u)) \to A(u)$ in the weak sense (along line segments). Taking the limit:

$$\langle A(u), w - u\rangle \geq 0 \quad \forall w \in K,$$

so u solves (4.1). This completes the proof [3, 5]. □

This reformulation is particularly valuable because it shifts the focus from evaluating A at the solution u to testing A at arbitrary points $v \in K$. As shown in Section 4.1.2, this equivalence facilitates the development of computational methods, such as fixed-point iterations, by leveraging the projection operator and the structure of monotone operators [1].

Computational Implementation

Fixed-Point Formulation For maximal monotone operators $A : H^1(\Omega) \to H^{-1}(\Omega)$, Minty's reformulation (Theorem 4.1.1) enables a fixed-point formulation that is computationally tractable. Define the operator:

$$G(w) = w - A(w). \tag{4.3}$$

The VI solution corresponds to finding $u \in K$ such that:

$$u = P_K(u - A(u)), \tag{4.4}$$

where $P_K : H^1(\Omega) \to K$ is the projection onto the closed, convex set K.

To justify this equivalence, we prove that (4.4) is equivalent to the VI (4.1), leveraging Minty's reformulation [3].

Justification of Fixed-Point Equivalence. Let $H = H^1(\Omega)$, a Hilbert space, and $K \subset H$ a nonempty, closed, convex subset. The projection $P_K(z)$ satisfies the variational inequality:

$$\langle z - P_K(z), v - P_K(z) \rangle \leq 0 \quad \forall v \in K,$$

since $P_K(z)$ is the point in K closest to z.

(1) VI $\implies$ Fixed-Point: Suppose $u \in K$ solves (4.1), i.e., $\langle A(u), v-u \rangle \geq 0$ for all $v \in K$. We need to show $u = P_K(u - A(u))$. By definition, $u = P_K(z)$ if and only if:

$$\langle z - u, v - u \rangle \leq 0 \quad \forall v \in K.$$

Set $z = u - A(u)$. Then:

$$\langle (u - A(u)) - u, v - u \rangle = \langle -A(u), v - u \rangle \leq 0,$$

since $\langle A(u), v - u \rangle \geq 0$ by (4.1). Thus, $u = P_K(u - A(u))$.

(2) Fixed-Point $\implies$ VI: Suppose $u = P_K(u - A(u))$, i.e.:

$$\langle (u - A(u)) - u, v - u \rangle \leq 0 \quad \forall v \in K.$$

This simplifies to:

$$\langle -A(u), v - u \rangle \leq 0 \implies \langle A(u), v - u \rangle \geq 0 \quad \forall v \in K,$$

which is exactly (4.1). Thus, u solves the VI.

Alternatively, using Minty's reformulation, suppose $u = P_K(u - A(u))$. By Theorem 4.1.1, the VI is equivalent to $\langle A(v), v - u\rangle \geq 0$ for all $v \in K$. Since $u = P_K(u - A(u))$, we have:

$$\langle (u - A(u)) - u, v - u\rangle \leq 0 \implies \langle -A(u), v - u\rangle \leq 0.$$

Now, consider Minty's condition. For any $v \in K$, since A is monotone:

$$\langle A(v) - A(u), v - u\rangle \geq 0 \implies \langle A(v), v - u\rangle \geq \langle A(u), v - u\rangle \geq 0,$$

where the last inequality follows from $\langle A(u), v - u\rangle \geq 0$. Thus, u satisfies Minty's condition, hence the VI (4.1) [5, 1]. □

This fixed-point formulation is a direct consequence of Minty's reformulation, as it exploits the equivalence between the VI and the condition involving arbitrary v. The projection P_K ensures that the solution remains in K, making (4.4) a natural basis for computational algorithms [3].

Numerical Methods Equation (4.4) enables several computational approaches for solving the VI:

- **Proximal algorithms**: These leverage the proximal operator, related to P_K, to iteratively approach the solution [1].
- **Fixed-point iterations**: Using a step size $\alpha > 0$, iterate:

  $$u_{n+1} = P_K(u_n - \alpha A(u_n)).$$

  Convergence is guaranteed for small α when A is monotone and Lipschitz continuous [5].
- **Variational discretization techniques**: Finite element methods discretize K and A, solving the fixed-point problem (4.4) on a finite-dimensional subspace [3].

These methods are particularly effective in applications like American option pricing, where the VI models the free boundary, as discussed in Chapter 3 [1].

Projection Properties The projection operator $P_K : H^1(\Omega) \to K$ is well-defined since K is closed and convex in the Hilbert space $H^1(\Omega)$:

$$P_K(z) = \arg\min_{y \in K} \|z - y\|^2. \tag{4.5}$$

Key properties include:

- **Non-expansivity**: $\|P_K(z_1) - P_K(z_2)\| \leq \|z_1 - z_2\|$, ensuring stability.
- **Characterization**: $u = P_K(z)$ if and only if:

  $$\langle z - u, v - u\rangle \leq 0 \quad \forall v \in K.$$

These properties, detailed in [5], ensure that the fixed-point problem (4.4) is well-posed and computationally feasible, especially when combined with Minty's reformulation [2].

Hence, we can state that variational inequalities are both mathematically rigorous and computationally tractable for real-world applications.

4.1.3 Numerical Methods for Parabolic VIs

The pricing of American put options, formulated as a parabolic variational inequality (PVI) in previous chapter, requires robust numerical methods to approximate the solution in practical applications. The PVI (3.14), derived from the Black-Scholes-Merton (BSM) formulation under risk-neutral pricing, is given by:

$$\min\left\{\frac{\partial V}{\partial q} + AV - f, V - \psi\right\} = 0, \tag{4.6}$$

where:

- $V(q, Y_q)$ is the option value at time-to-expiry $q = T - t$ with underlying asset price Y_q,
- $AV = -\frac{1}{2}\sigma^2 Y_q^2 \frac{\partial^2 V}{\partial Y_q^2} - rY_q \frac{\partial V}{\partial Y_q} + rV$ is the elliptic operator, with r the risk-free rate and σ the volatility,
- $\psi(Y_q) = \max(K - Y_q, 0)$ is the payoff for strike price K,
- $f(Y_q) = 0$ (no running payoff),
- The solution lies in $L^2(0, T; H^1(\Omega)) \cap H^1(0, T; L^2(\Omega))$, with initial condition $V(0, Y_q) = \psi(Y_q)$.

This PVI captures the optimal stopping problem of American options, with $\frac{\partial V}{\partial q} + AV = 0$ in the continuation region ($V > \psi$) and $V = \psi$ in the exercise region, separated by a non-smooth free boundary Y_q^* . The computational challenges include discretizing the infinite-dimensional Sobolev-Bochner space, handling the non-smooth free boundary, and enforcing the constraint $V \geq \psi$.

To address these issues, we explore numerical methods from [1] : the penalization and finite difference methods combined, and the projection method. The penalization approach transforms the PVI into a smoother nonlinear PDE, which is then discretized using finite difference schemes for numerical solution. Projection methods leverage Minty's reformulation to enforce constraints directly. Each method is detailed below, with its general theory, adaptation to the American put PVI, and a theoretical pseudocode outlining the computational process.

Penalization and Finite Difference Methods

General Theory The penalization method, as outlined in [1] , approximates a VI by introducing a penalty term that softly enforces the constraint, converting the inequality into a nonlinear PDE. For a general PVI:

$$\min\left\{\frac{\partial u}{\partial t} + Au - f, u - \psi\right\} = 0,$$

with $u \in K = \{v \in H^1(\Omega) \mid v \geq \psi\}$, the penalized problem is:

$$\frac{\partial u_\varepsilon}{\partial t} + Au_\varepsilon + \frac{1}{\varepsilon}\beta(u_\varepsilon - \psi) = f,$$

where $\varepsilon > 0$ is a small parameter, and $\beta(s) = -s^- = \min(s, 0)$ penalizes violations of $u_\varepsilon \geq \psi$. As $\varepsilon \to 0$, $u_\varepsilon \to u$ in appropriate function spaces (e.g., $L^2(0, T; H^1(\Omega))$), as shown before. The resulting PDE benefits from elliptic regularity, making it amenable to numerical discretization methods.

Finite difference schemes, detailed in [1] , discretize such PDEs over a grid in time and space, approximating derivatives with difference quotients. The Crank-Nicolson scheme, which balances explicit and implicit methods for stability, is applied to the penalized PDE, with the penalty term evaluated at the new time level:

$$\frac{u^{n+1} - u^n}{\Delta t} + \frac{1}{2}\left[Au^{n+1} + Au^n\right] + \frac{1}{\varepsilon}\beta(u^{n+1} - \psi) = f^n.$$

The penalty must be treated implicitly: with an explicit penalty $\beta(u^n - \psi)$, the scheme is unstable unless Δt is of the order of ε. The resulting nonlinear system is solved at each time step by Newton (policy) iteration, each iteration being a tridiagonal linear system [11]. This combination leverages the smoothness of the penalized PDE with the stability and efficiency of finite differences.

Adaptation to American Put PVI For the American put PVI (4.6), with $f = 0$, the penalized PDE becomes:

$$\frac{\partial V_\varepsilon}{\partial q} + AV_\varepsilon + \frac{1}{\varepsilon}\beta(V_\varepsilon - \psi) = 0,$$

where:

$$AV_\varepsilon = -\frac{1}{2}\sigma^2 Y_q^2 \frac{\partial^2 V_\varepsilon}{\partial Y_q^2} - rY_q \frac{\partial V_\varepsilon}{\partial Y_q} + rV_\varepsilon,$$

$\psi(Y_q) = \max(K - Y_q, 0)$, and $\beta(V_\varepsilon - \psi) = \min(V_\varepsilon - \psi, 0)$. The initial condition is $V_\varepsilon(0, Y_q) = \psi(Y_q)$, and boundary conditions are $V_\varepsilon(q, 0) = K$ (deep in-the-money) and $V_\varepsilon(q, Y_{\max}) \to 0$ (far out-of-the-money).

Using a finite difference grid with $q_n = n\Delta q$, $\Delta q = T/N$, and $Y_m = m\Delta Y$, $\Delta Y = Y_{\max}/M$, the Crank-Nicolson scheme at (q_n, Y_m) is:

$$\frac{V_\varepsilon^{n+1,m} - V_\varepsilon^{n,m}}{\Delta q} + \frac{1}{2}\left[AV_\varepsilon^{n+1,m} + AV_\varepsilon^{n,m}\right] + \frac{1}{\varepsilon}\beta(V_\varepsilon^{n+1,m} - \psi_m) = 0,$$

where:

$$AV_\varepsilon^{n,m} = -\frac{1}{2}\sigma^2 Y_m^2 \frac{V_\varepsilon^{n,m+1} - 2V_\varepsilon^{n,m} + V_\varepsilon^{n,m-1}}{\Delta Y^2} - rY_m \frac{V_\varepsilon^{n,m+1} - V_\varepsilon^{n,m-1}}{2\Delta Y} + rV_\varepsilon^{n,m},$$

and $\psi_m = \max(K - Y_m, 0)$. At each time step this is solved by Newton (policy) iteration: given the current iterate, the penalty is switched on at the nodes where $V_\varepsilon^{n+1,m} < \psi_m$, which gives a tridiagonal system, solvable using efficient algorithms like the Thomas algorithm; the iteration stops when the set of penalized nodes no longer changes [11]. The penalty term $\frac{1}{\varepsilon}\beta(V_\varepsilon^{n+1,m} - \psi_m)$ ensures $V_\varepsilon \geq \psi$, with convergence to V as $\varepsilon \to 0$.

Theoretical Pseudocode The combined penalization and finite difference process is:

1. **Initialize**: Set ε, $\Delta q = T/N$, $\Delta Y = Y_{\max}/M$, grid points $q_n = n\Delta q$, $Y_m = m\Delta Y$, and $V_\varepsilon^{0,m} = \max(K - Y_m, 0)$.
2. **Set Boundary Conditions**: For all n, set $V_\varepsilon^{n,0} = K$, $V_\varepsilon^{n,M} = 0$.
3. **Time Loop**: For each $n = 0, \ldots, N-1$:
   (a) Compute $AV_\varepsilon^{n,m}$ and $AV_\varepsilon^{n+1,m}$ using central differences for spatial derivatives.
   (b) Newton (policy) iteration: switch the penalty $\frac{1}{\varepsilon}\beta(V_\varepsilon^{n+1,m} - \psi_m)$ on at the nodes where the current iterate lies below ψ_m.
   (c) Form and solve the Crank-Nicolson tridiagonal system for $V_\varepsilon^{n+1,m}$; repeat (b)–(c) until the set of penalized nodes no longer changes.
4. **Output**: Return $V_\varepsilon^{N,m}$ as the approximate option value at $q = T$.
5. **Refine**: Decrease ε, Δq, ΔY to ensure convergence to V.

Projection Methods

General Theory Projection methods, described in [1] , solve VIs by reformulating them as fixed-point problems using projections onto the constraint set, often leveraging Minty's reformulation. For a VI:

$$\langle \frac{\partial u}{\partial t} + Au - f, v - u \rangle \geq 0 \quad \forall v \in K,$$

the solution satisfies:

$$u = P_K\left(u - \alpha\left(\frac{\partial u}{\partial t} + Au - f\right)\right),$$

where P_K is the projection onto K, and $\alpha > 0$ is a step size. For $K = \{v \in H^1(\Omega) \mid v \geq \psi\}$, $P_K(z) = \max(z, \psi)$. This leads to iterative schemes like proximal algorithms, which converge under monotonicity and coercivity.

Adaptation to American Put PVI For the PVI (4.6), we use the projection method with Minty's reformulation. The VI is equivalent to finding $V \in K = \{v \in H^1(\Omega) \mid v \geq \psi\}$ such that:

$$V = P_K \left(V - \alpha \left(\frac{\partial V}{\partial q} + AV \right) \right),$$

where $P_K(V) = \max(V, \psi(Y_q))$, and $AV = -\frac{1}{2}\sigma^2 Y_q^2 \frac{\partial^2 V}{\partial Y_q^2} - rY_q \frac{\partial V}{\partial Y_q} + rV$. We discretize this using finite differences for derivatives and iterate to find V. Initial condition: $V(0, Y_q) = \psi(Y_q)$. Boundary conditions: $V(q, 0) = K$, $V(q, Y_{\max}) = 0$. The projection ensures $V \geq \psi$, and convergence is supported by the Lions-Stampacchia Theorem.

Theoretical Pseudocode The projection method process is:

1. **Initialize**: Set step size α, $\Delta q = T/N$, $\Delta Y = Y_{\max}/M$, grid points q_n, Y_m, and $V^{0,m} = \max(K - Y_m, 0)$.
2. **Set Boundary Conditions**: For all n, set $V^{n,0} = K$, $V^{n,M} = 0$.
3. **Time Loop**: For each $n = 0, \ldots, N-1$:
   (a) Initialize iteration: Set $V^{n+1,m,0} = V^{n,m}$.
   (b) Iteration Loop: For $k = 0, \ldots, K_{\max}$:
      i. Compute $\frac{\partial V^{n+1,m,k}}{\partial q} \approx \frac{V^{n+1,m,k} - V^{n,m}}{\Delta q}$.
      ii. Compute $AV^{n+1,m,k}$ using central differences.
      iii. Update: $V^{n+1,m,k+1} = \max\left(V^{n+1,m,k} - \alpha \left(\frac{\partial V^{n+1,m,k}}{\partial q} + AV^{n+1,m,k} \right), \psi_m \right)$.
   (c) Set $V^{n+1,m} = V^{n+1,m,K_{\max}}$.
4. **Output**: Return $V^{N,m}$ as the approximate option value.
5. **Refine**: Adjust α, Δq, ΔY, $K_{\max}$ for convergence.

Conclusion

The numerical methods,penalization with finite differences and projection, offer distinct approaches to solving the American put PVI (4.6), each with strengths and limitations:

- **Penalization and Finite Difference Methods**:
  - *Pros*: Converts the VI to a smoother PDE, leveraging stable finite difference schemes (e.g., Crank-Nicolson); supports regularity analysis ; efficient for low-dimensional problems.
  - *Cons*: Requires small ε, increasing computational cost; penalty term may introduce stiffness; grid-based, leading to high cost in high dimensions with coarse Δq, ΔY.

- **Projection Methods**:
  - *Pros*: Directly enforces $V \geq \psi$ via projections; computationally efficient for simple constraints; leverages Minty's reformulation.
  - *Cons*: Iterative nature may converge slowly; requires careful choice of step size α.

These methods provide a robust foundation for pricing American options, building on the theoretical framework of Chapters presented above. The penalization and finite difference approach excels in structured settings with smooth approximations, while projection methods offer flexibility for constraint enforcement.

Projected Successive Over-Relaxation

After discretization in space and implicit time stepping, the American put variational inequality can be written, at each time level, as a linear complementarity problem. In one consistent sign convention, the discrete unknown $\mathbf{V} \in \mathbb{R}^m$ satisfies

$$A\mathbf{V} - \mathbf{b} \geq 0, \qquad \mathbf{V} - \boldsymbol{\psi} \geq 0, \qquad (\mathbf{V} - \boldsymbol{\psi})^{\mathsf{T}}(A\mathbf{V} - \mathbf{b}) = 0, \tag{4.7}$$

where A is the matrix produced by the time–space discretization, $\mathbf{b}$ contains known terms from the preceding time level, and $\boldsymbol{\psi}$ is the discretized payoff. Inequalities in (4.7) are interpreted componentwise. The complementary condition means that, at each node, either the discrete pricing equation is satisfied or the obstacle is active.

Projected successive over-relaxation (PSOR) adapts the SOR iteration to this complementarity constraint. For a current iterate $\mathbf{V}^{(k)}$, first form the componentwise Gauss-Seidel row update

$$\widehat{V}_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j<i} a_{ij} V_j^{(k+1)} - \sum_{j>i} a_{ij} V_j^{(k)} \right), \qquad i = 1, \ldots, m. \tag{4.8}$$

The relaxed value is then projected onto the obstacle:

$$V_i^{(k+1)} = \max \left\{ \psi_i,\, V_i^{(k)} + \omega \big(\widehat{V}_i^{(k+1)} - V_i^{(k)} \big) \right\}. \tag{4.9}$$

The relaxation parameter ω controls the size of each SOR update. The iteration sweeps through the spatial nodes, repeating (4.8)–(4.9) until a specified complementarity residual or iterate-difference tolerance is met. It is then advanced to the next time level. Projection enforces $V_i \geq \psi_i$ at every iteration; this distinguishes PSOR from an unconstrained SOR solution followed by a single projection at the end. In the implementation used here, the relaxation parameter is selected according to the Young's optimal ω scheme, described in the accompanying code. The method is classical for complementarity problems and American option valuation[12].

Sign-convention note. If the finite-difference system in the implementation is written with the opposite inequality or with $A\mathbf{V}+\mathbf{b}$ instead of $A\mathbf{V}-\mathbf{b}$, rewrite the linear complementarity system and the row update consistently.

4.1.4 Conclusion

The computational framework for variational inequalities, as developed in this section, transforms theoretical constructs into practical tools for applications such as American option pricing. Minty's reformulation provides an equivalent characterization of suitable monotone variational inequalities and helps motivate computational approaches based on projection and fixed-point iterations, as well as finite-element discretizations [3, 5]. For parabolic VIs, penalization combined with finite-difference methods and projection-based approaches address the challenges posed by obstacle constraints and non-smooth free boundaries [1]. Penalization methods, implemented with implicit Crank–Nicolson schemes, approximate the constrained problem through a penalized PDE, but small penalty parameters can increase computational cost and numerical stiffness. Alternatively, projected successive over-relaxation (PSOR) methods solve the discrete linear complementarity problem by combining successive over-relaxation with projection onto the payoff constraint. This enforces the obstacle constraint at each iteration and provides a practical iterative approach to American option pricing [12, 13]. The efficiency of PSOR depends on the discretized system, the relaxation parameter, and the convergence tolerance. Together, penalization, projection, and PSOR illustrate complementary approaches to solving variational inequalities, balancing constraint enforcement, numerical accuracy, and computational efficiency.

Chapter 5

Physics-Informed Neural Networks for Parabolic Variational Inequalities

5.1 Introduction

The numerical methods explored previously, penalization, finite differences, and projection methods, provide robust solutions to the parabolic variational inequality (PVI) for pricing American put options, as formulated above. However, these grid-based approaches face challenges in high-dimensional settings and with non-smooth free boundaries, which are inherent to optimal stopping problems. Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. [7], offer a promising alternative by leveraging deep learning to approximate solutions in a mesh-free, data-driven framework. Originally designed for partial differential equations (PDEs), PINNs have been extended to variational inequalities (VIs), making them suitable for the American put PVI [9].

This chapter provides a comprehensive exploration of PINNs, detailing their theoretical foundations, adaptation to the PVI, and computational implementation. We begin with a general theory of PINNs, emphasizing their neural network architecture, loss function design, and training process, particularly for inequalities. We then adapt PINNs to the American put PVI, incorporating the constraint $V \geq \psi$ and handling non-smoothness via viscosity solutions . A theoretical pseudocode outlines the computational process, and the conclusion evaluates the pros and cons of PINNs compared to classical methods, setting the stage for future comparisons.

5.2 Physics-Informed Neural Networks

5.2.1 General Theory

Physics-Informed Neural Networks (PINNs) are a class of deep learning models that embed physical laws, expressed as PDEs or VIs, directly into the neural network's loss function, enabling the approximation of solutions without extensive labeled data [6, 7]. Unlike traditional numerical methods that discretize the domain, PINNs represent the solution as a neural network $u_\theta(t, x)$, parameterized by weights and biases θ, and train it to satisfy the governing equations and boundary conditions over a set of collocation points.

Neural Network Architecture

A PINN typically employs a feedforward neural network with multiple hidden layers, where each neuron computes:

$$z^{(l+1)} = \varphi(W^{(l)} z^{(l)} + b^{(l)}),$$

with $W^{(l)}$, $b^{(l)}$ as weights and biases, and φ a nonlinear activation function (e.g., tanh, ReLU). For a PVI, the input is (t, x) (or (q, Y_q) in our context), and the output is the solution $u_\theta(t, x)$. The network's flexibility allows it to approximate complex, non-smooth functions, crucial for VIs with free boundaries.

Loss Function for PDEs

For a PDE of the form:

$$\frac{\partial u}{\partial t} + \mathcal{N}[u] = f, \quad u(0, x) = u_0(x), \quad u(t, \partial\Omega) = g(x),$$

the PINN loss function combines residuals of the PDE, initial condition, and boundary conditions:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{PDE}} + \mathcal{L}_{\text{initial}} + \mathcal{L}_{\text{boundary}},$$

where:

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N_r} \sum_{i=1}^{N_r} \left| \frac{\partial u_\theta}{\partial t}(t_i, x_i) + \mathcal{N}[u_\theta](t_i, x_i) - f(t_i, x_i) \right|^2,$$

$$\mathcal{L}_{\text{initial}} = \frac{1}{N_0} \sum_{i=1}^{N_0} \left| u_\theta(0, x_i) - u_0(x_i) \right|^2,$$

$$\mathcal{L}_{\text{boundary}} = \frac{1}{N_b} \sum_{i=1}^{N_b} \left| u_\theta(t_i, x_i) - g(x_i) \right|^2.$$

Here, $\{(t_i, x_i)\}_{i=1}^{N_r}$, $\{x_i\}_{i=1}^{N_0}$, and $\{(t_i, x_i)\}_{i=1}^{N_b}$ are collocation points sampled in the domain, initial condition, and boundary, respectively. Automatic differentiation computes derivatives $\frac{\partial u_\theta}{\partial t}$, ∇u_θ, etc., enabling gradient-based optimization [6].

Extending PINNs to Variational Inequalities

For VIs, such as:

$$\min\left\{\frac{\partial u}{\partial t} + Au - f, u - \psi\right\} = 0,$$

PINNs must enforce the inequality constraint $u \geq \psi$ and the complementarity condition. Combettes and Pesquet [9] propose deep neural network structures for VIs, suggesting penalty-based approaches to incorporate constraints. A suitable loss function for PINNs is:

$$\mathcal{L}_{\text{VI}} = \frac{1}{N_r}\sum_{i=1}^{N_r}\left(\left|\frac{\partial u_\theta}{\partial t} + Au_\theta - f\right|^2 \mathbf{1}_{\{u_\theta > \psi\}} + |u_\theta - \psi|^2 \mathbf{1}_{\{u_\theta \leq \psi\}}\right),$$

where $\mathbf{1}$ is the indicator function, approximated by a smooth sigmoid for differentiability [8]. This enforces the VI by ensuring the PDE holds in the continuation region ($u_\theta > \psi$) and $u_\theta = \psi$ in the constrained region. Additionally, a constraint penalty ensures $u_\theta \geq \psi$:

$$\mathcal{L}_{\text{constraint}} = \frac{1}{N_r}\sum_{i=1}^{N_r}\left[\min(0, u_\theta(t_i, x_i) - \psi(t_i, x_i))\right]^2.$$

The total loss is:

$$\mathcal{L}(\theta) = w_1\mathcal{L}_{\text{VI}} + w_2\mathcal{L}_{\text{constraint}} + w_3\mathcal{L}_{\text{initial}} + w_4\mathcal{L}_{\text{boundary}},$$

with weights w_i balancing terms [6]. This approach aligns with the penalization method but operates in a mesh-free setting.

Training Process

The network is trained by minimizing $\mathcal{L}(\theta)$ using stochastic gradient descent (SGD) or advanced optimizers (e.g., Adam). Collocation points are sampled randomly or via Latin hypercube sampling to cover the domain [7]. The training process leverages automatic differentiation for PDE/VI residuals and iterates until convergence, monitored by the loss value or validation metrics. PINNs are particularly effective for non-smooth solutions, as they do not require grid-based discretization and can capture free boundaries implicitly.

Challenges and Enhancements

PINNs face challenges, including high computational cost, sensitivity to hyperparameter tuning (e.g., network depth, learning rate), and balancing loss terms. Recent advancements include adaptive sampling [9] and residual-based attention mechanisms [8] to focus training on high-error regions. For VIs, ensuring the constraint $u \geq \psi$ is critical, often requiring careful weight tuning or architectural modifications (e.g., positive output layers).

5.3 Adaptation to American Put PVI

5.3.1 Formulation

American options differ from European options in that they can be exercised at any time before expiration. This early exercise feature introduces a free boundary problem, making the valuation mathematically more challenging.

We consider the American put option with strike price K, volatility σ, risk-free rate r, and time to maturity T. The pricing function $u(S,t)$, representing the value of the option at asset price S and time t, satisfies the following conditions:

- **Payoff condition:** $u(S,T) = \Phi(S) = \max(K - S, 0)$,
- **Inequality constraint:** $u(S,t) \geq \Phi(S)$,
- **PDE constraint:** $\frac{\partial u}{\partial t} + \mathcal{L}u - ru \leq 0$,
- **Complementarity condition:**

$$(u(S,t) - \Phi(S)) \cdot \left(\frac{\partial u}{\partial t} + \mathcal{L}u - ru \right) = 0,$$

  ensuring that either the PDE holds or the option value equals the payoff.

Here, $\mathcal{L}$ is the standard Black-Scholes differential operator:

$$\mathcal{L}u = \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 u}{\partial S^2} + rS\frac{\partial u}{\partial S}.$$

5.3.2 PINN Architecture and Boundary Conditioning

We implement a physics-informed neural network (PINN) to approximate the solution $u(S,t)$ using a fully connected deep neural network. The network takes as input the pair (S,t) and outputs a scalar value representing the option price.

To satisfy the boundary condition $u(0,t) = K$ exactly, we apply a boundary conditioning technique to the network output. Specifically, the output $u(S,t)$ is modeled as:

$$u_\theta(S,t) = Ke^{-S/S_{\max}} + \left(1 - e^{-S/S_{\max}}\right) \cdot f_\theta(S,t),$$

where f_θ is the raw neural network output. This structure enforces $u_\theta(0,t) = K$ for every t without any constraint in the loss; the far-field condition $u(S_{\max}, t) \approx 0$ and the terminal payoff are imposed through the loss terms below.

5.3.3 Loss Function Based on the Variational Inequality Residual

Let $u_\theta(S,t)$ denote the PINN approximation to the American put value on $S \in [0, S_{\max}]$ and $t \in [0,T]$. Define the calendar-time Black–Scholes residual by

$$\mathcal{B}[u_\theta](S,t) = \frac{\partial u_\theta}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 u_\theta}{\partial S^2} + rS\frac{\partial u_\theta}{\partial S} - ru_\theta, \qquad \Phi(S) = (K - S)^+. \quad (5.1)$$

In calendar time, the American put variational inequality is expressed through the complementarity condition

$$\min\{-\mathcal{B}[u],\, u-\Phi\}=0. \tag{5.2}$$

The continuation region satisfies $\mathcal{B}[u]=0$ and $u>\Phi$, while in the exercise region $u=\Phi$ and the Black–Scholes equation need not hold as an equality of zero residual. The interior loss therefore penalizes the residual of the variational inequality directly, rather than imposing the Black–Scholes PDE at every interior point:

$$L_{\mathrm{VI}}(\theta)=\frac{1}{N_{\mathrm{int}}}\sum_{i=1}^{N_{\mathrm{int}}}\left[\min\{-\mathcal{B}[u_\theta](S_i,t_i),u_\theta(S_i,t_i)-\Phi(S_i)\}\right]^2. \tag{5.3}$$

Here (S_i,t_i) are interior collocation points. This loss encodes the continuation equation, the obstacle inequality, and their complementarity in a single residual term.

The output transformation

$$u_\theta(S,t)=Ke^{-S/S_{\max}}+\left(1-e^{-S/S_{\max}}\right)f_\theta(S,t) \tag{5.4}$$

where f_θ is the raw network output, enforces $u_\theta(0,t)=K$ exactly. The far-field boundary and terminal payoff are imposed by separate losses:

$$L_{\mathrm{term}}(\theta)=\frac{1}{N_T}\sum_{j=1}^{N_T}\left[u_\theta(S_j,T)-\Phi(S_j)\right]^2, \tag{5.5}$$

$$L_{\mathrm{far}}(\theta)=\frac{1}{N_B}\sum_{j=1}^{N_B}u_\theta(S_{\max},t_j)^2. \tag{5.6}$$

The total training objective is

$$L(\theta)=L_{\mathrm{VI}}(\theta)+\lambda_{\mathrm{term}}L_{\mathrm{term}}(\theta)+\lambda_{\mathrm{far}}L_{\mathrm{far}}(\theta), \tag{5.7}$$

where λ_{term} and λ_{far} are the terminal and far-boundary weights used in the executable implementation.

This formulation avoids forcing $\mathcal{B}[u_\theta]=0$ in the exercise region. That distinction is material: for an American put with $S<K$, the payoff is $\Phi(S)=K-S$ and $\mathcal{B}[\Phi]=-rK$ away from the payoff kink. A loss that independently penalizes $\mathcal{B}[u_\theta]^2$ throughout the domain can therefore oppose the obstacle condition in the exercise region. The VI-residual formulation in (5.3) avoids this particular conflict.

5.3.4 Sampling Strategy and Training Details

The implementation uses a fully connected network with four hidden layers, 64 units per hidden layer, and the hyperbolic tangent activation function. Training

is carried out with Adam for 20,000 iterations. Interior collocation points are sampled over $S \in [0, S_{\max}]$ and $t \in [0, T]$, with denser sampling near the strike and near the ends of the time interval, where the solution changes most sharply. Terminal points are sampled at $t = T$, and far-boundary points are sampled at $S = S_{\max}$. The boundary at $S = 0$ is enforced exactly by (5.4). [1]

5.3.5 Training Algorithm

1. Initialize the four-hidden-layer network f_θ (64 units per layer, tanh activation) and use (5.4) to obtain u_θ.

2. Sample interior collocation points (S_i, t_i), terminal points S_j at $t = T$, and far-boundary times t_j at $S = S_{\max}$. Bias interior sampling toward the strike and toward $t = 0$ and $t = T$ as specified by the implementation.

3. Use automatic differentiation to evaluate u_t, u_S, and u_{SS}, and form $\mathcal{B}[u_\theta]$ using (5.1).

4. Compute L_{VI}, L_{term}, and L_{far} from (5.3)–(5.6), combine them according to (5.7), and update θ with Adam.

5. Repeat for 20,000 iterations, recording losses and diagnostics. Return the trained function $u_\theta(S, t)$ as the approximation to the American put value.

Benchmark losses. The earlier weighted loss can be retained in the computational study as a baseline, together with the reweighted variant, but it should not be described as the primary loss used by the current implementation. The baseline and the VI-residual model differ in the interior residual construction; the terminal and boundary treatment should otherwise be held fixed when making a controlled comparison.

[1] The weights λ_{term} and λ_{far}, the number of collocation points per batch and the learning-rate schedule are tuning choices and were not optimised. The values used for the reported results are set in the repository, where they can be changed to reproduce or extend the experiments.

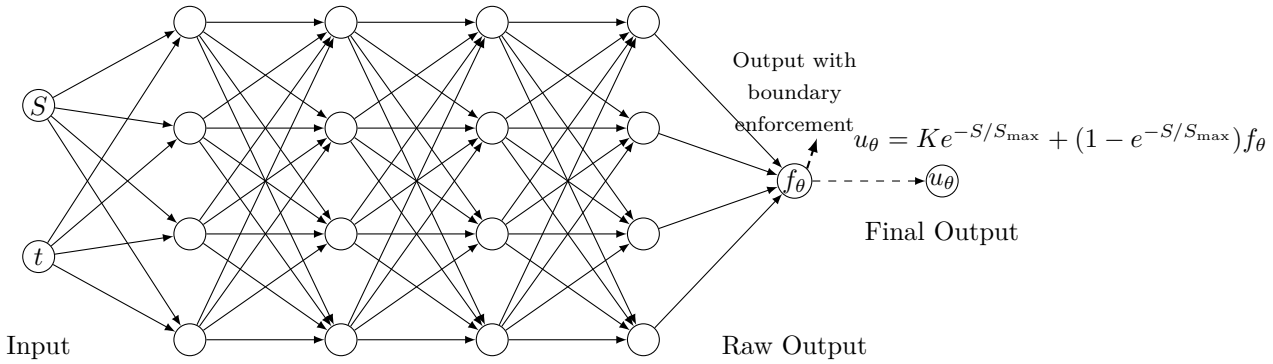


Figure 5.1: Neural network architecture used in the PINN: a fully connected feedforward model with four hidden layers and an output transformation to satisfy boundary conditions.

5.4 Conclusion

Physics-Informed Neural Networks offer a powerful, mesh-free approach to solving the American put PVI, complementing the classical methods of Chapter 4. Their strengths and limitations are:

- **Pros**:
  - Mesh-free, scalable to high-dimensional problems (e.g., multi-asset options).
  - Implicitly captures non-smooth free boundaries, aligning with viscosity solutions (p. 27).
  - Flexible loss function design accommodates VIs via penalties or soft constraints [9].
- **Cons**:
  - High computational cost and training time, especially for deep networks [6].
  - Sensitive to hyperparameter tuning (e.g., w_i, learning rate), requiring experimentation.
  - Convergence not guaranteed without adaptive strategies [8].

PINNs leverage the theoretical framework, particularly the Sobolev-Bochner spaces and viscosity solutions.

Appendix

A1. Proof of the Lions-Stampacchia Theorem

Proof of the Lions–Stampacchia theorem

Proof. By the Riesz representation theorem, there exist a bounded linear operator $A : H \to H$ and an element $g \in H$ such that

$$(Au, v)_H = a(u, v), \qquad (g, v)_H = \langle f, v \rangle_{H^*,H} \quad \text{for all } u, v \in H.$$

The continuity and coercivity assumptions imply

$$\|Au - Av\|_H \le M \|u - v\|_H , \qquad (Au - Av, u - v)_H \ge \alpha \|u - v\|_H^2 .$$

Let $P_K : H \to K$ denote the metric projection onto the nonempty closed convex set K. It is non-expansive and satisfies

$$u = P_K(z) \quad \Longleftrightarrow \quad (z - u, v - u)_H \le 0 \quad \text{for all } v \in K.$$

For $\rho > 0$, define $T_\rho : K \to K$ by

$$T_\rho(u) = P_K\big(u - \rho(Au - g)\big).$$

A point $u \in K$ is a fixed point of T_ρ if and only if

$$(-\rho(Au - g), v - u)_H \le 0 \quad \text{for all } v \in K,$$

which is equivalent to

$$(Au - g, v - u)_H \ge 0 \quad \text{for all } v \in K.$$

By the definitions of A and g, this is exactly (1.3).

It remains to prove that T_ρ has a unique fixed point. For $u, v \in K$, non-expansiveness of P_K gives

$$\begin{aligned} \|T_\rho(u) - T_\rho(v)\|_H^2 &\le \|(u - v) - \rho A(u - v)\|_H^2 \\ &= \|u - v\|_H^2 - 2\rho(A(u - v), u - v)_H + \rho^2 \|A(u - v)\|_H^2 \\ &\le \big(1 - 2\rho\alpha + \rho^2 M^2\big) \|u - v\|_H^2 . \end{aligned}$$

Choose ρ so that $0 < \rho < 2\alpha/M^2$. Since $\alpha \le M$, this gives $0 \le 1-2\rho\alpha+\rho^2M^2 < 1$, so T_ρ is a contraction on K. The set K is complete in the metric induced by H, because it is closed. The Banach fixed-point theorem therefore gives a unique fixed point $u \in K$. By the fixed-point equivalence already established, this u is the unique solution of (1.3). □

Coercivity reduction for the monotone-operator extension

Proof using the bounded-set existence theorem. We use the abstract bounded-set existence theorem for monotone, hemicontinuous operators from Kinderlehrer and Stampacchia, Chapter III, Section 1[3]. For $R > 0$, define

$$K_R = K \cap \overline{B}(x_0, R).$$

For all sufficiently large R, this set is nonempty, bounded, closed, and convex. The cited bounded-set theorem gives a point $x_R \in K_R$ such that

$$\langle F(x_R), y - x_R\rangle \ge 0 \qquad \text{for all } y \in K_R.$$

Choosing $y = x_0$ yields

$$\langle F(x_R), x_R - x_0\rangle \le 0.$$

By coercivity, this is impossible when $\|x_R - x_0\| = R$ and R is sufficiently large. Hence, for such an R, x_R lies strictly inside the ball $B(x_0, R)$.

Now take any $y \in K$. Since x_R is interior to the ball and K is convex, there exists $t \in (0, 1]$ sufficiently small that $y_t = x_R + t(y - x_R) \in K_R$. The truncated variational inequality implies

$$0 \le \langle F(x_R), y_t - x_R\rangle = t\langle F(x_R), y - x_R\rangle.$$

Dividing by $t > 0$ gives

$$\langle F(x_R), y - x_R\rangle \ge 0 \qquad \text{for all } y \in K,$$

so x_R solves the original variational inequality on K.

For uniqueness, suppose $x_1, x_2 \in K$ are both solutions. Testing the inequality for x_1 with $y = x_2$ and the inequality for x_2 with $y = x_1$, then adding, gives

$$\langle F(x_1) - F(x_2), x_1 - x_2\rangle \le 0.$$

Strict monotonicity makes the left-hand side positive whenever $x_1 \ne x_2$; therefore $x_1 = x_2$. □

Scope of this proof. The coercivity reduction above is explicit, but it invokes the bounded-set abstract existence theorem cited from Kinderlehrer–Stampacchia. It does not reproduce the book's proof of that auxiliary theorem. The preceding contraction argument, by contrast, is a complete standalone proof of the classical Lions–Stampacchia theorem for continuous coercive bilinear forms.

A2. A toy example for the future

Consider two countries, $\mathcal{A}$ and $\mathcal{B}$, engaged in a high-tension diplomatic standoff over a contested region. Country $\mathcal{A}$ is considering whether to:

(i) Escalate the situation through sanctions or military exercises;

(ii) De-escalate by making concessions;

(iii) Delay action and monitor developments.

Let $V(t, x)$ represent the expected *strategic value* of the situation for country $\mathcal{A}$ at time t, given state variable x, which encodes intelligence metrics (e.g., troop movement, public sentiment, alliance strength).

Dynamics

Assume that x_t evolves according to a controlled stochastic differential equation:

$$dx_t = \mu(x_t, a_t)dt + \sigma(x_t)dW_t,$$

where $a_t \in \{0, 1\}$ is the decision ($0 =$ wait, $1 =$ act), and W_t is a standard Wiener process modeling uncertainty (e.g., unexpected moves by $\mathcal{B}$ or international intervention).

Payoff and Constraint Structure

At any time t, country $\mathcal{A}$ receives an instantaneous payoff:

$$f(x_t) = \text{geopolitical gain from maintaining status quo at } x_t.$$

However, country $\mathcal{A}$ faces a constraint: *it cannot allow the perceived threat level to exceed a tolerance threshold $\psi(x_t)$ without responding.* Hence, its decision must satisfy:

$$V(t, x) \geq \psi(x), \quad \text{(security constraint)}.$$

This reflects political, military, and public pressure to act when the situation crosses a red line.

Variational Inequality Formulation

The optimal decision problem becomes a parabolic variational inequality:

$$\min\left\{-\frac{\partial V}{\partial t} - \mathcal{L}V - f(x),\ V - \psi(x)\right\} = 0, \quad \text{in } [0,T]\times\mathbb{R},$$

where $\mathcal{L}$ is the infinitesimal generator:

$$\mathcal{L}V = \mu(x,a)\frac{\partial V}{\partial x} + \frac{1}{2}\sigma^2(x)\frac{\partial^2 V}{\partial x^2}.$$

The constraint $V \geq \psi(x)$ enforces the rule: once the threat level exceeds tolerance, waiting is no longer feasible—forcing an immediate escalation or concession.

Interpretation

This problem captures the essence of a diplomatic dilemma:

- Wait and gather more information, hoping the threat subsides (but risk acting too late),
- Act prematurely, potentially overreacting to noise,
- Operate under hard constraints, such as alliance obligations or red-line policies.

This toy model can be extended to include game-theoretic interactions (e.g., $\mathcal{B}$ optimizing a parallel value function), and solved numerically using methods analogous to those in American option pricing.

Bibliography

[1] A. E. Tsekrekos and A. N. Yannacopoulos, *Variational Inequalities in Management Science and Finance: Modelling, Analysis, Numerics and Applications*, EURO Advanced Tutorials on Operational Research, Springer, 2024.

[2] J.-L. Lions and G. Stampacchia, *Variational Inequalities*, Communications on Pure and Applied Mathematics, vol. 20, pp. 493–519, 1967.

[3] D. Kinderlehrer and G. Stampacchia, *An Introduction to Variational Inequalities and Their Applications*, Academic Press, 1980.

[4] M. G. Crandall, H. Ishii, and P.-L. Lions, *User's Guide to Viscosity Solutions of Second Order Partial Differential Equations*, Bulletin of the American Mathematical Society, vol. 27, pp. 1–67, 1992.

[5] H. Brezis, *Functional Analysis, Sobolev Spaces and Partial Differential Equations*, Springer, 2011.

[6] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, *Physics-informed machine learning*, Nature Reviews Physics, vol. 3, pp. 422–440, 2021.

[7] M. Raissi, P. Perdikaris, and G. E. Karniadakis, *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations*, Journal of Computational Physics, vol. 378, pp. 686–707, 2019.

[8] J. Sirignano and K. Spiliopoulos, *DGM: A deep learning algorithm for solving partial differential equations*, Journal of Computational Physics, vol. 375, pp. 1339–1364, 2018.

[9] P. L. Combettes and J.-C. Pesquet, *Deep Neural Network Structures Solving Variational Inequalities*, Set-Valued and Variational Analysis, vol. 28, pp. 491–518, 2020.

[10] J. Simon, *Compact sets in the space $L^p(0,T;B)$*, Annali di Matematica Pura ed Applicata, vol. 146, pp. 65–96, 1987.

[11] P. A. Forsyth and K. R. Vetzal, *Quadratic convergence for valuing American options using a penalty method*, SIAM Journal on Scientific Computing, vol. 23, no. 6, pp. 2095–2122, 2002.

[12] C. W. Cryer, "The solution of a quadratic programming problem using systematic overrelaxation," *SIAM Journal on Control*, vol. 9, no. 3, pp. 385–392, 1971. doi:10.1137/0309028

[13] M. A. H. Dempster and J. P. Hutton, "Pricing American Stock Options by Linear Programming," *Mathematical Finance*, vol. 9, no. 3, pp. 229–254, 1999. doi: 10.1111/1467-9965.00069.